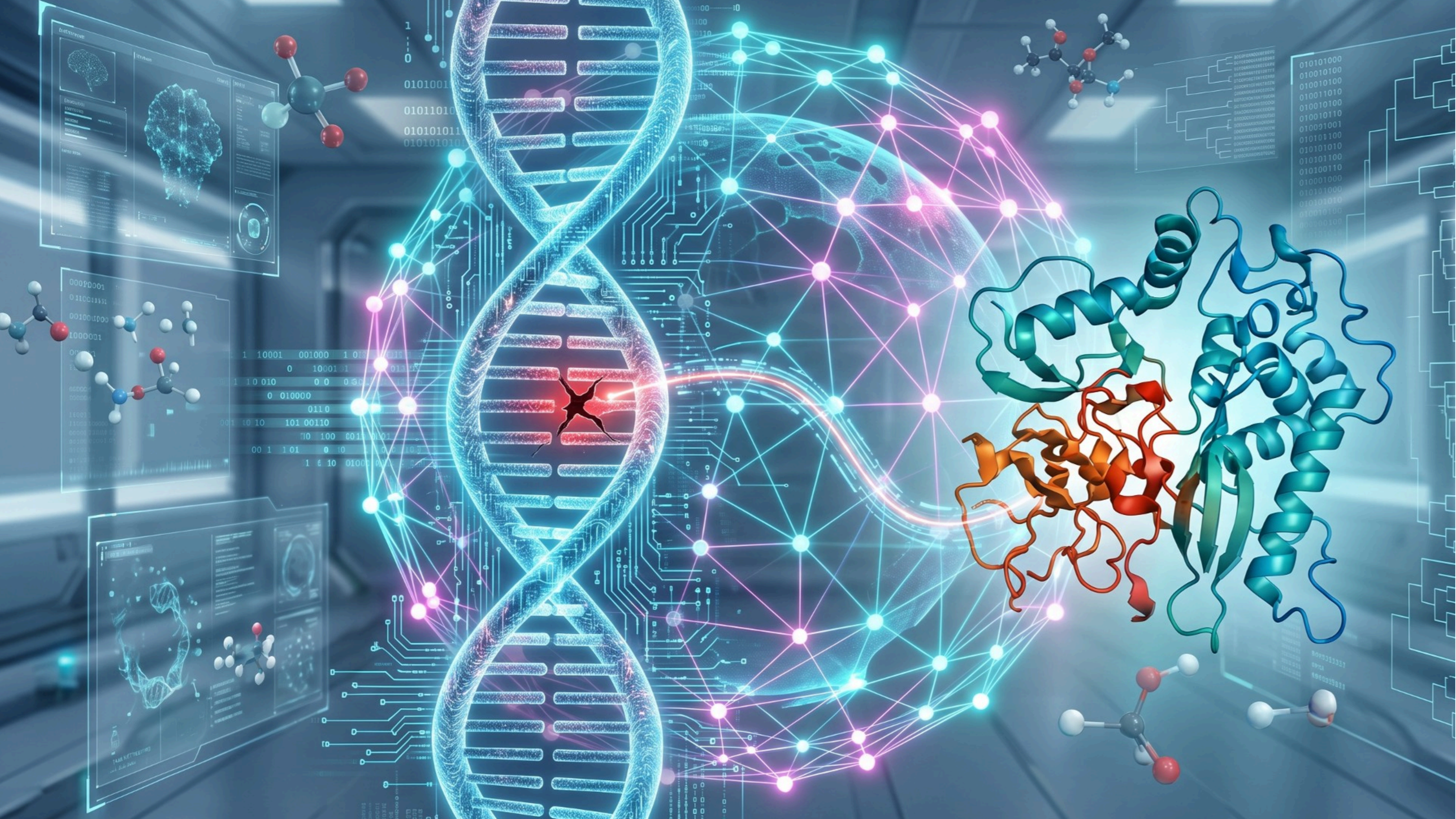
Using AI to Visualize the Causes of Disease



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1 The Central Dogma (Flow of Genetic Information)

2 Reproducing the Central Dogma in Python

3 Why Protein Structure Matters, and AlphaFold

4 Applications in Medicine and Pharmacology

Contents

1 The Central Dogma (Flow of Genetic Information)

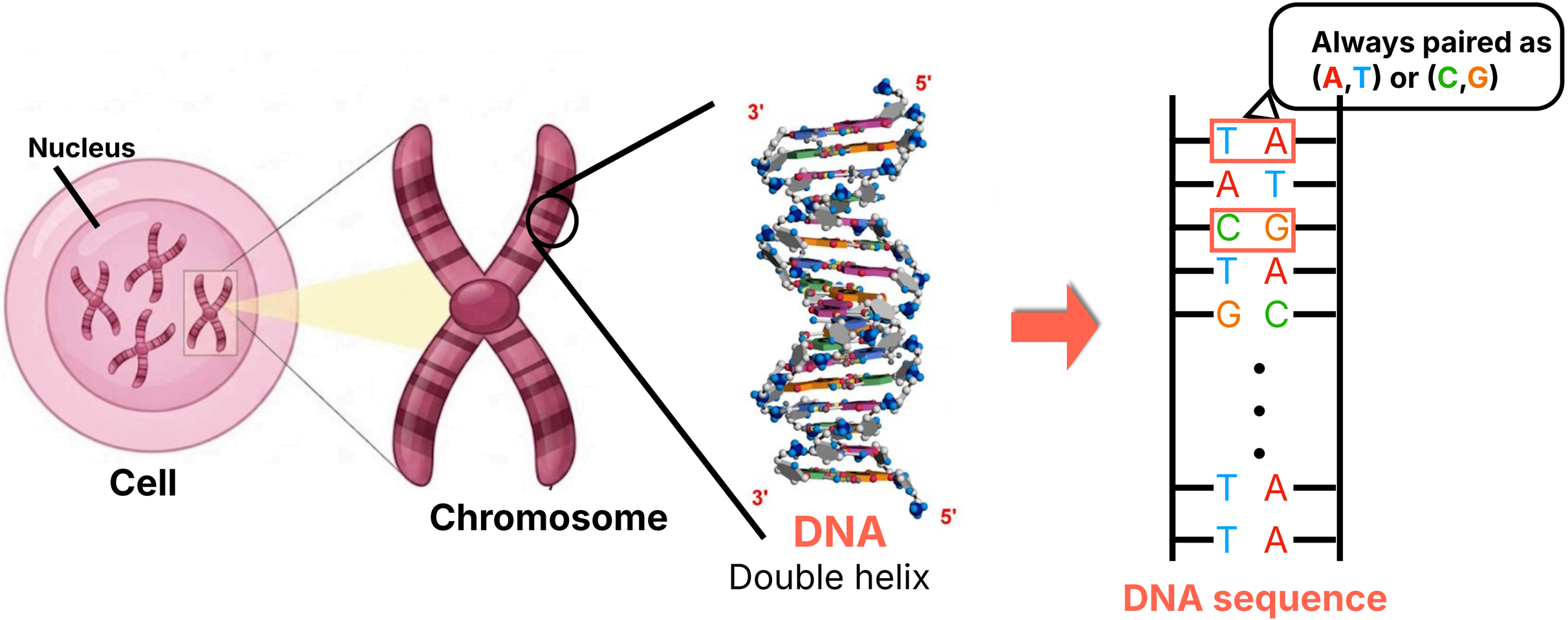
2 Reproducing the Central Dogma in Python

3 Why Protein Structure Matters, and AlphaFold

4 Applications in Medicine and Pharmacology

DNA: The Blueprint of the Body

DNA stores the genetic information that shapes life as sequences of 4 kinds of letters

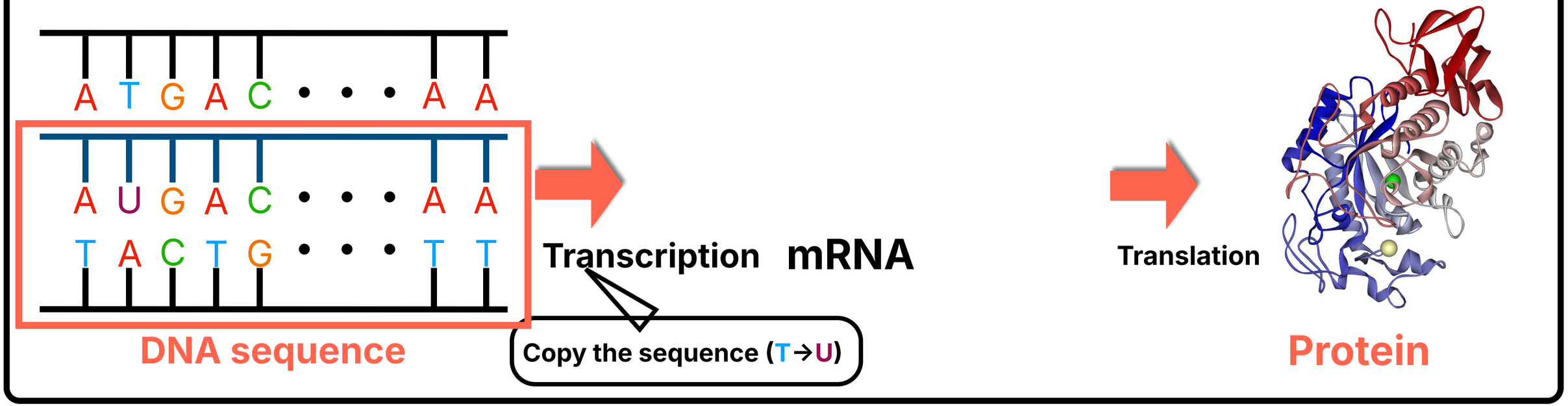


Question How does this simple string of letters build our bodies?

The Central Dogma

Question How does this simple string of letters build our bodies?

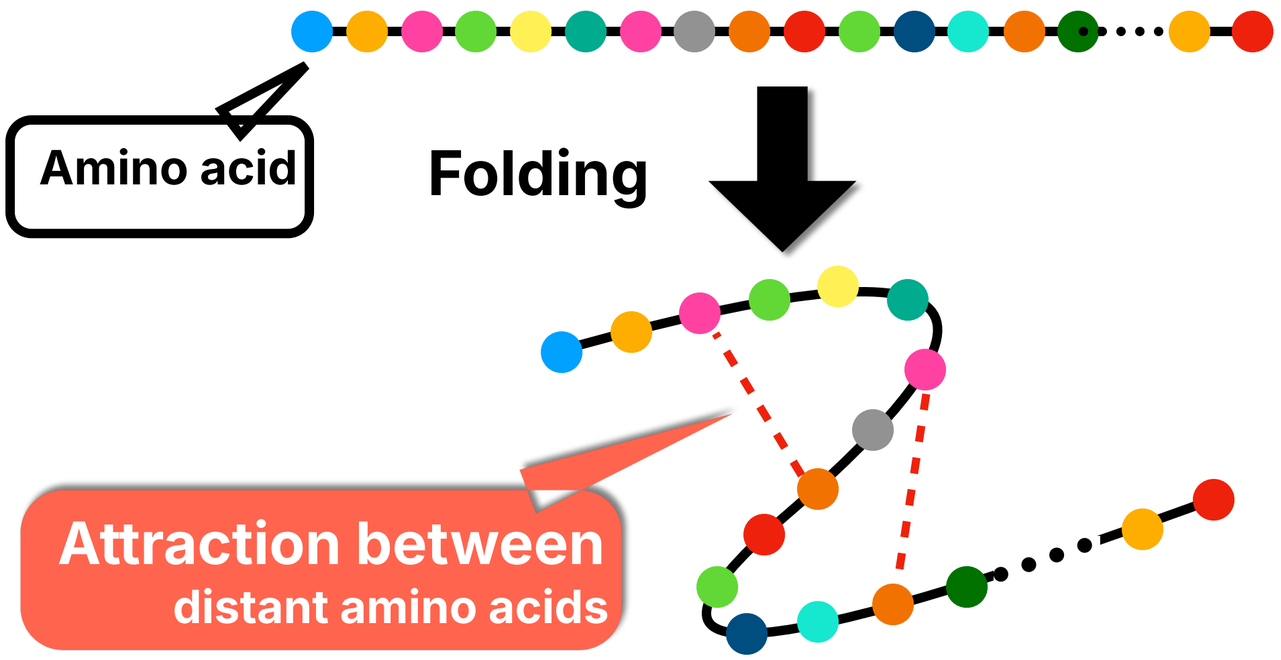
The Central Dogma (Flow of Genetic Information)



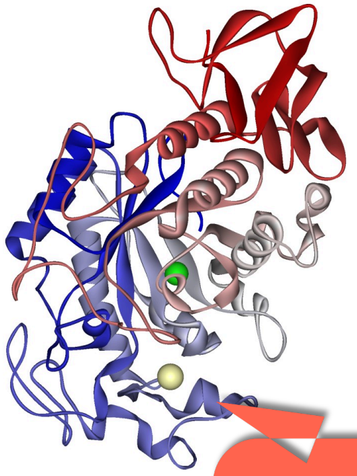
Answer From the DNA sequence, molecules called proteins, which have diverse functions and shape life, are made

Protein

Amino acids: molecules (20 kinds) that link up in chains of **tens to thousands**
The type and order of amino acids determine how the chain folds and what it does



Further Folding



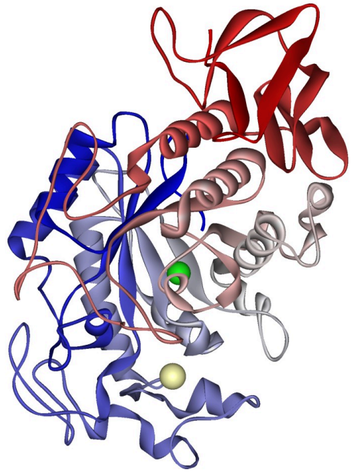
When folded correctly, it becomes functional

Some amino acids are attracted to distant amino acids in the chain
As a result, the amino acid chain folds into a functional protein

Diverse Proteins

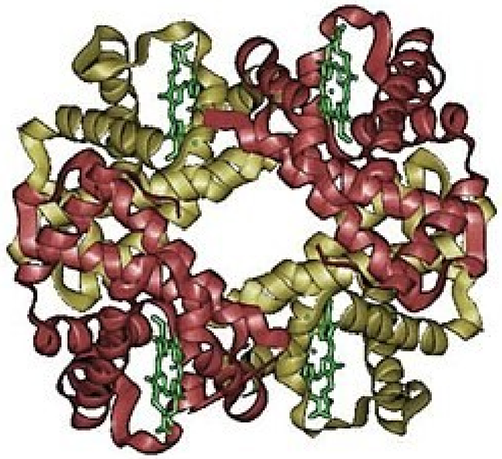
Depending on amino acid type and order, chains fold into diverse shapes with diverse functions

Amylase



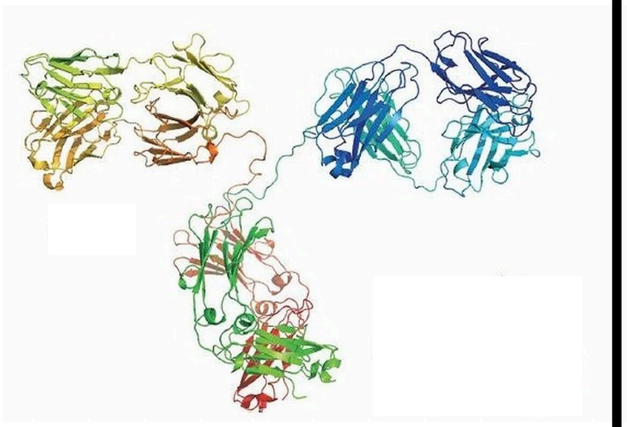
A **digestive enzyme** in saliva
Breaks down starch (digestion)

Hemoglobin



Found in red blood cells
Carries oxygen throughout the body

Antibody



Recognizes and destroys invading viruses (immunity)

Diverse Proteins

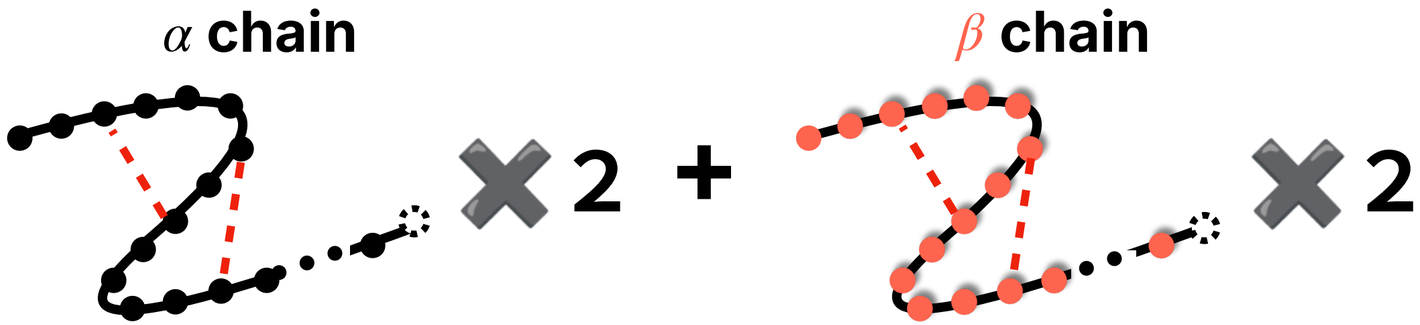
Hemoglobin

Contained in red blood cells, and carries oxygen throughout the body

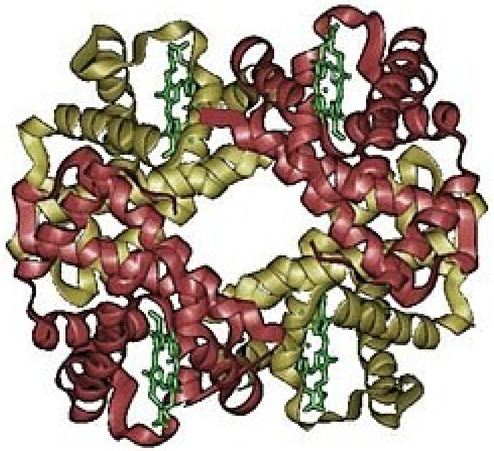
A protein made of 4 amino acid chains assembled together

Binds oxygen in the lungs
Releases it in tissues

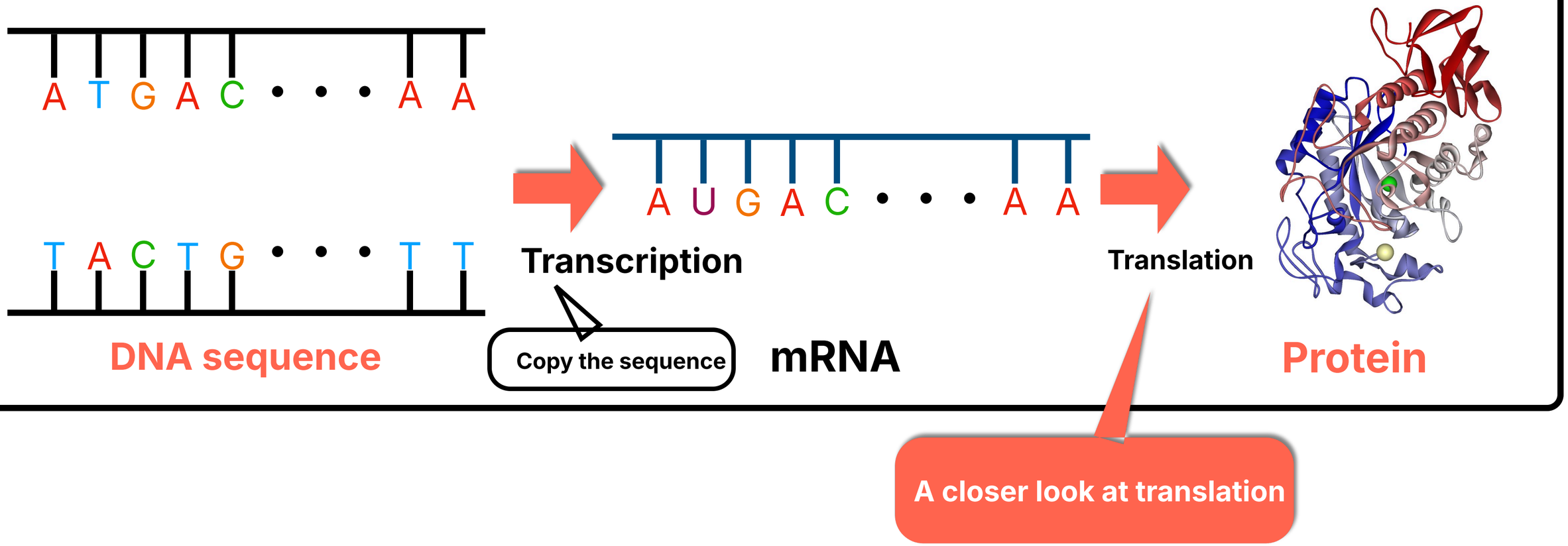




Hemoglobin

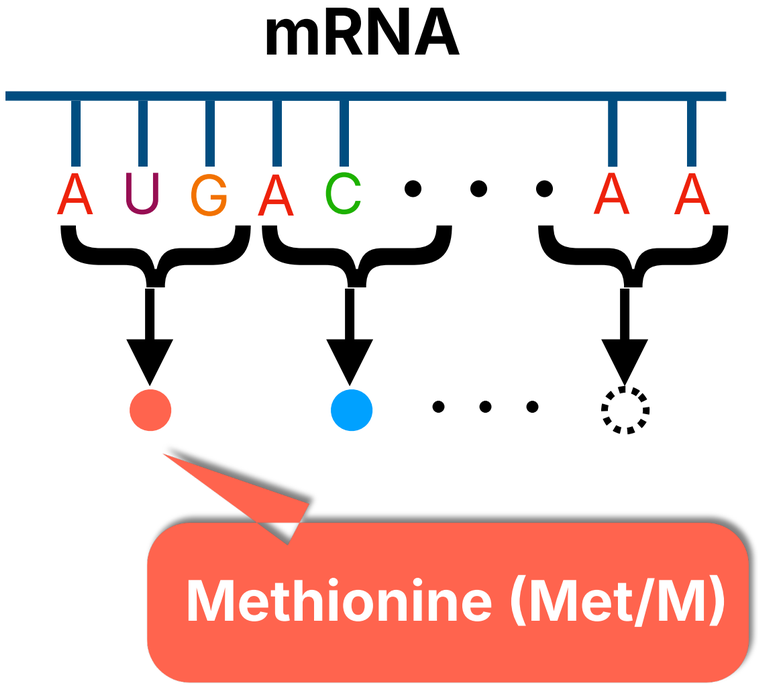


The Central Dogma (Flow of Genetic Information)

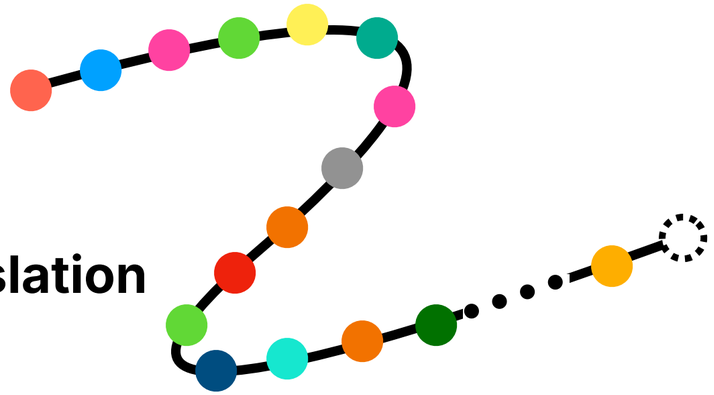


Translation

Every 3 letters of mRNA specify one amino acid, building an amino acid chain



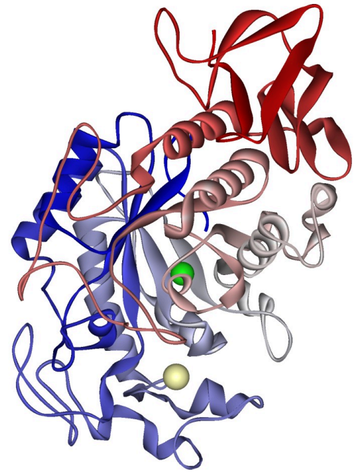
Translation



Amino acid chain

(Example: MVHLTPV・・・)

Folding

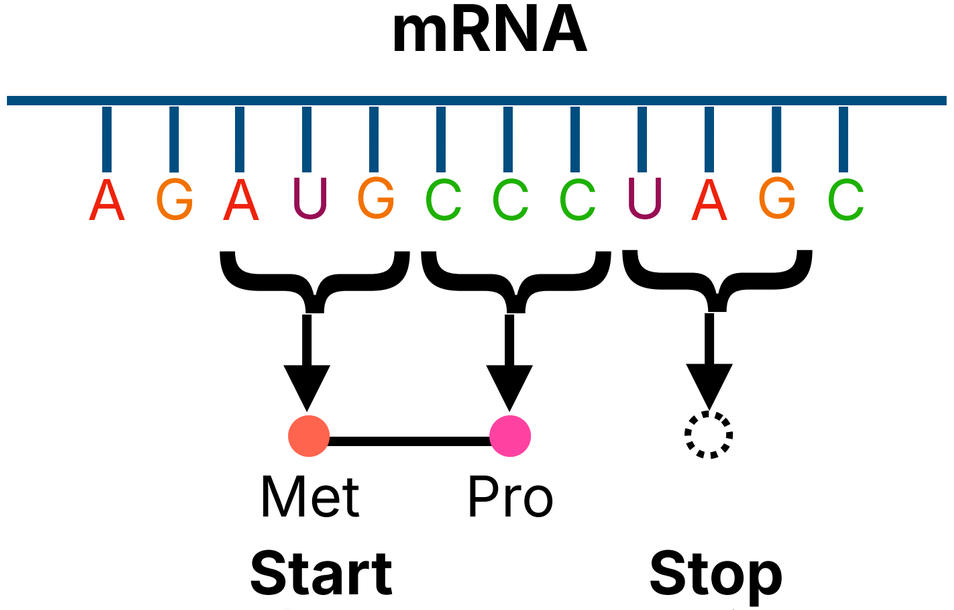


Protein

Which amino acid each 3-letter set of mRNA (codon) specifies is already known

Codon Table

1文字目		2文字目 U		2文字目 C		2文字目 A		2文字目 G	
U		UUU	フェニルアラニン Phe/F	UCU	セリン Ser/S	UAU	チロシン Tyr/Y	UGU	システイン Cyc/C
		UUC		UCC		UAC		UGC	
		UUA	ロイシン Leu/L	UCA		UAA	終止	UGA	終止
		UUG		UCG		UAG		UGC	トリプトファン Trp/W
C		CUU	ロイシン Leu/L	CCU	プロリン Pro/P	CAU	ヒスチジン His/H	CGU	アルギニン Arg/R
		CUC		CCC		CAC		CGC	
		CUA		CCA		CAA	グルタミン Gln/Q	CGA	
		CUG		CCG		CAG		CGG	
A		AUU	イソロイシン Ile/I	ACU	スレオニン Thr/T	AAU	アスパラギン Asn/N	AGU	セリン Ser/S
		AUC		ACC		AAC		AGC	
		AUA		ACA		AAA	リシン Lys/K	AGA	アルギニン Arg/R
		AUG	メチオニン(開始) Met/M	ACG		AAG		AGG	
G		GUU	バリン Val/V	GCU	アラニン Ala/A	GAU	アスパラギン酸 Asp/D	GGU	グリシン Gly/G
		GUC		GCC		GAC		GGC	
		GUA		GCA		GAA	グルタミン酸 Glu/E	GGA	
		GUG		GCG		GAG		GGG	

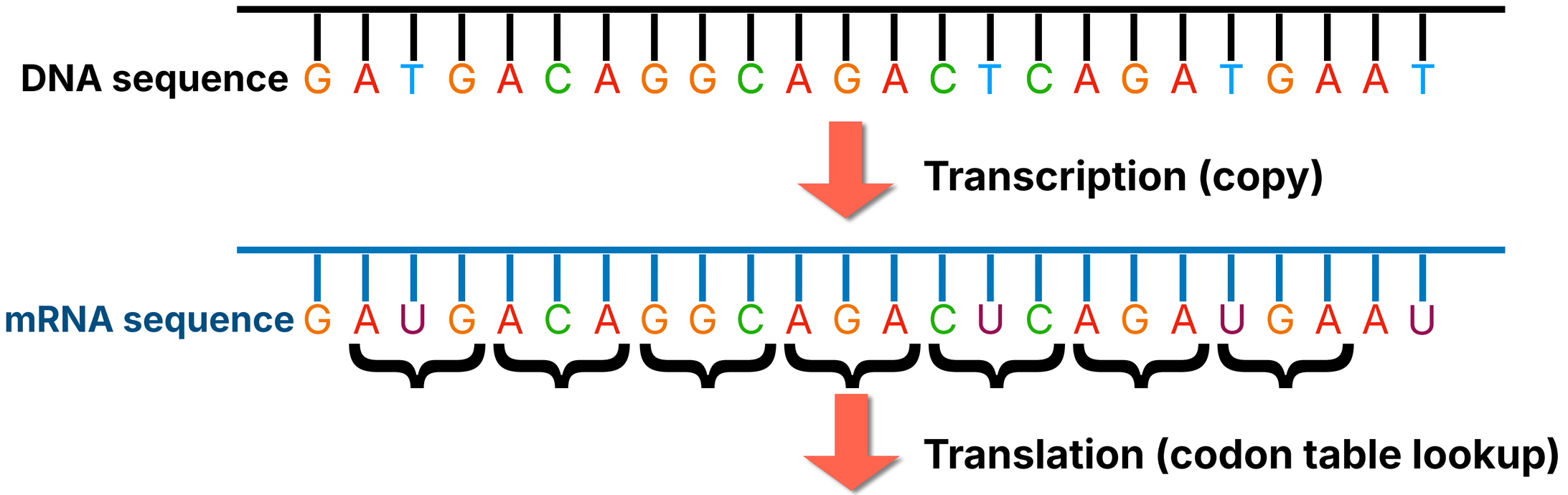


From “AUG” on the mRNA, translation starts

At a stop codon on the mRNA, translation ends

Summary

How DNA sequences become protein amino acid sequences is already understood: genes are expressed through simple sequence operations



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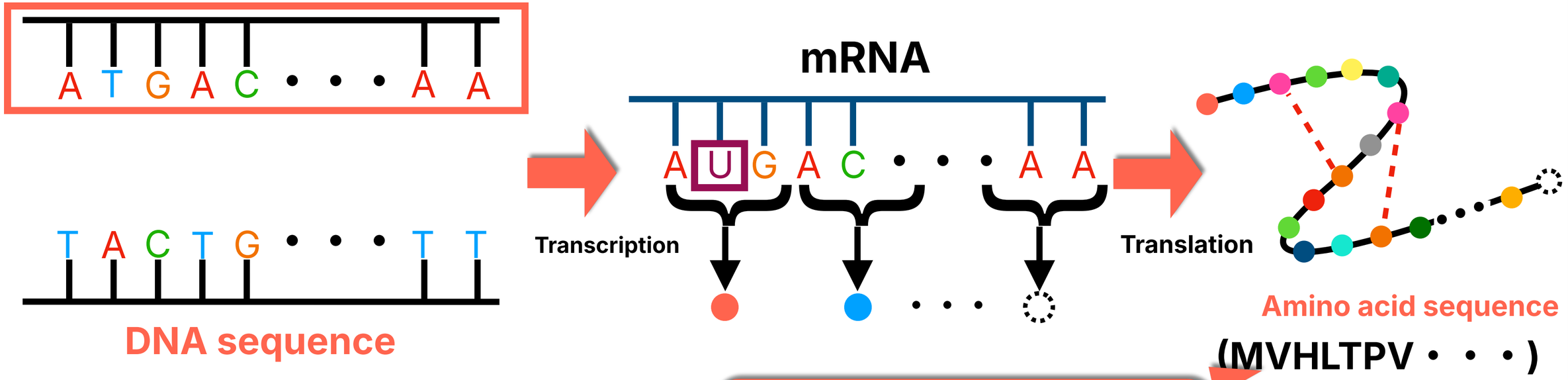
Python Exercise

Let's write a program that converts a DNA sequence into an amino acid sequence

The input is the DNA sense strand (it becomes mRNA when T → U)

Sense strand

The Central Dogma (Flow of Genetic Information)



Output amino acids as one-letter codes

Installing Google Colaboratory

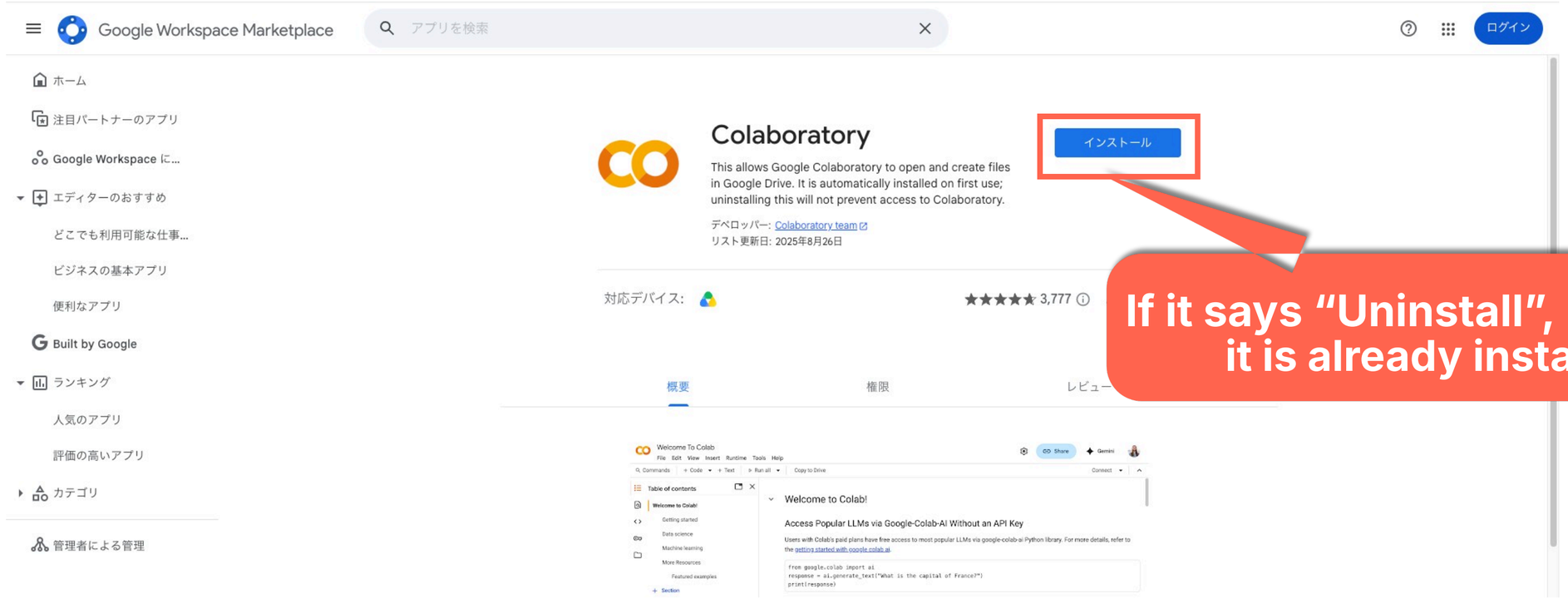
Google Colaboratory

A tool for writing and running Python code in the browser, with no setup required

If you haven't installed it yet, please install it

Use the link below to check whether it is installed, and to install it

Check Colaboratory installation

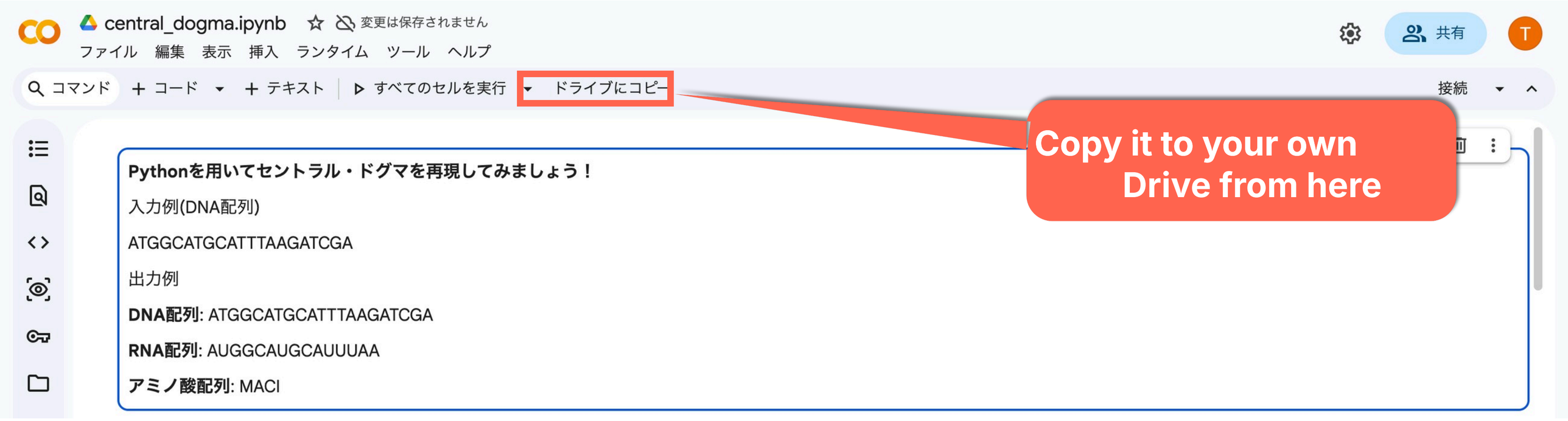


Template Files

Open a template file from the links below and copy it to your own Drive

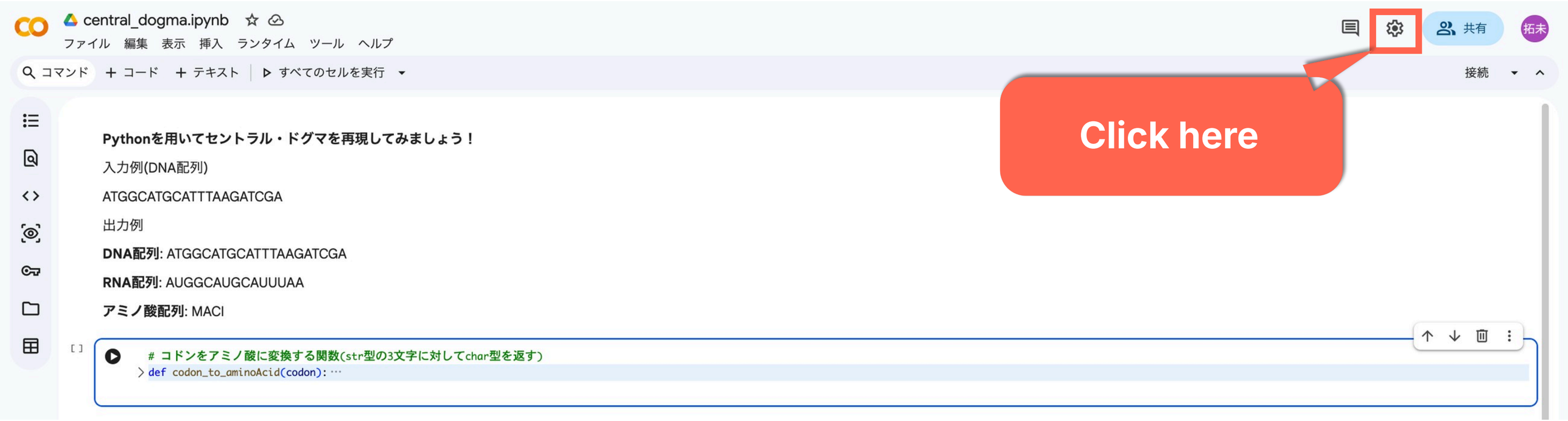
Two versions are provided to suit your Python level

Link: Template without hints ***Link: Template with hints***



Block AI Generation

We want you to deepen your grasp of the central dogma by coding it yourself, so please avoid using AI generation as much as possible



Block AI Generation

We want you to deepen your grasp of the central dogma by coding it yourself, so please avoid using AI generation as much as possible



- **No AI code completion**
- **Hide generative AI features**

Provided Function

Codon -> Amino acid

Given a 3-letter string of A, U, C, G as a codon, it returns the one-letter amino acid code
The codon_to_aminoAcid function is already provided, so feel free to use it

Argument: a 3-letter string (e.g., UAC, CAG)

```
# コドンをアミノ酸に変換する関数(str型の3文字に対してchar型を返す)
def codon_to_aminoAcid(codon):
    if len(codon) != 3:
      print("error : 与えられたコドン"+codon+"が3文字ではないため変換できません")
      return "?"
    # コドン表(コドンとそれがコードするアミノ酸(1文字表記)の対応関係)
    codon_table = {…
    }
    aminoAcid = codon_table.get(codon, "?")
    return aminoAcid
```

codon=“AUG”
aa = codon_to_aminoAcid(codon)
print(aa)

Output: M

Returns the one-letter amino acid code,
“*” for a stop codon, or “?” if it cannot be converted

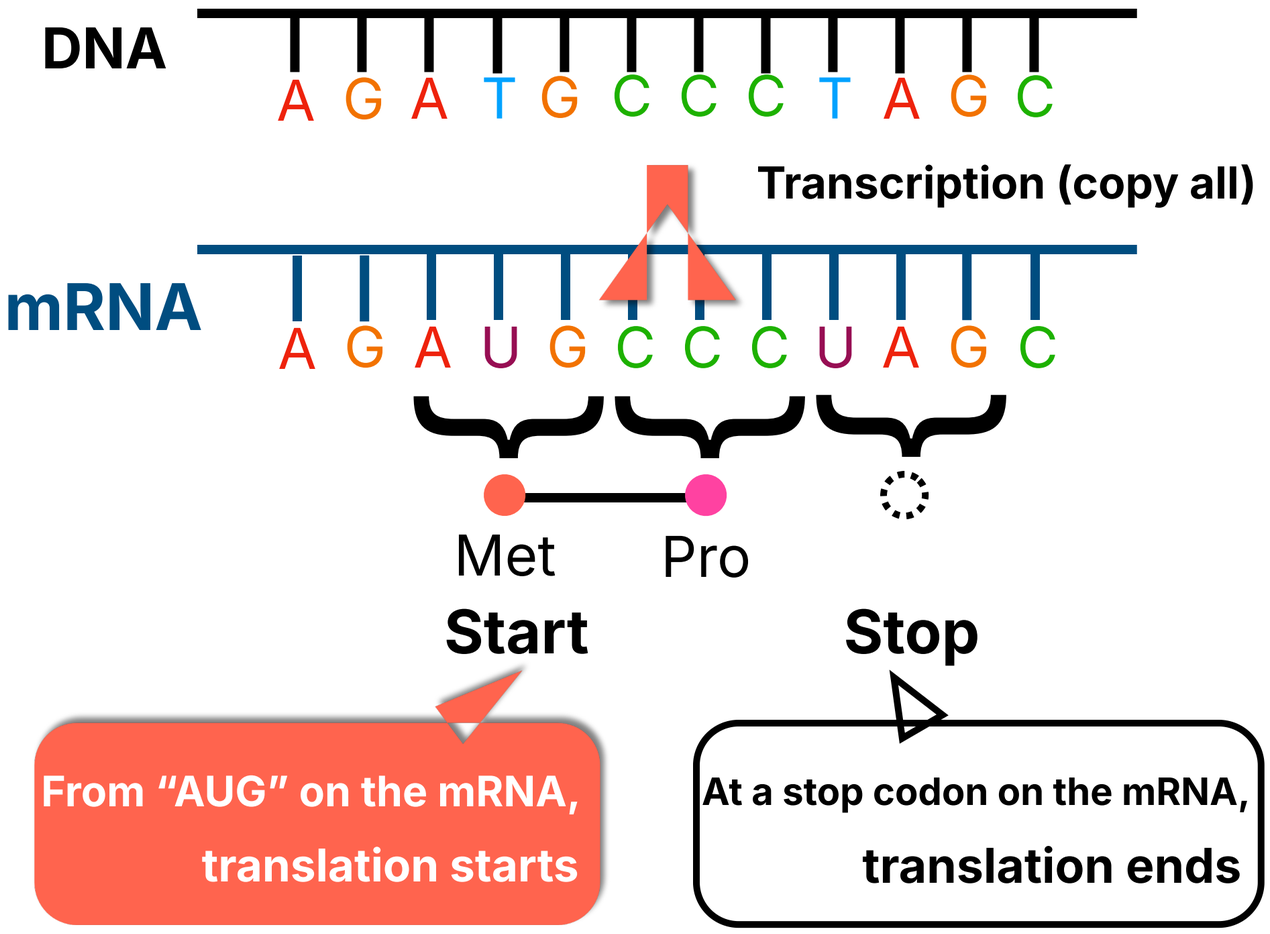
Notes

The mRNA sequence is not translated from end to end;

only the region from the start codon (AUG) to a stop codon (UAA, UAG, UGA) is translated

Codon Table

1文字目		2文字目 U		2文字目 C		2文字目 A		2文字目 G	
U		UUU	フェニルアラニン Phe/F	UCU	セリン Ser/S	UAU	チロシン Tyr/Y	UGU	システイン Cyc/C
		UUC		UCC		UAC		UGC	
		UUA	ロイシン Leu/L	UCA		UAA	終止	UGA	終止
		UUG		UCG		UAG		UGC	トリプトファン Trp/W
C		CUU	ロイシン Leu/L	CCU	プロリン Pro/P	CAU	ヒスチジン His/H	CGU	アルギニン Arg/R
		CUC		CCC		CAC		CGC	
		CUA		CCA		CAA	グルタミン Gln/Q	CGA	
		CUG		CCG		CAG		CGG	
A		AUU	イソロイシン Ile/I	ACU	スレオニン Thr/T	AAU	アスパラギン Asn/N	AGU	セリン Ser/S
		AUC		ACC		AAC		AGC	
		AUA		ACA		AAA	リシン Lys/K	AGA	アルギニン Arg/R
		AUG	メチオニン(開始) Met/M	ACG		AAG		AGG	
G		GUU	バリン Val/V	GCU	アラニン Ala/A	GAU	アスパラギン酸 Asp/D	GGU	グリシン Gly/G
		GUC		GCC		GAC		GGC	
		GUA		GCA		GAA	グルタミン酸 Glu/E	GGA	
		GUG		GCG		GAG		GGG	



Creating Amino Acid Sequences

Use your code to create amino acid sequences from the DNA sequences

1. DNA sequence of the hemoglobin α chain



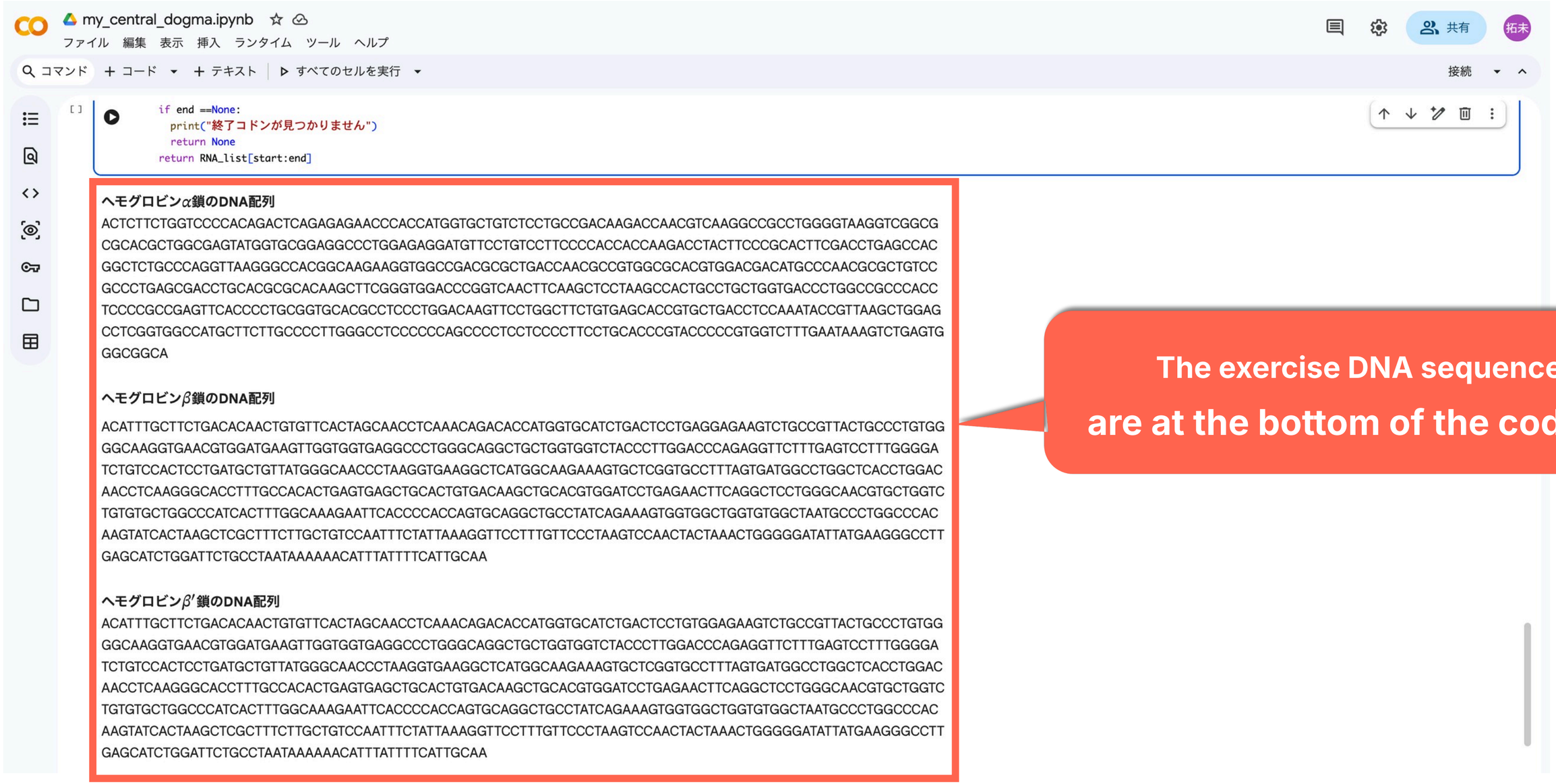
2. DNA sequence of the hemoglobin β chain



3. DNA sequence of the hemoglobin β' chain



Where the DNA Sequences Are



The exercise DNA sequences are at the bottom of the code

Checking Your Amino Acid Sequences

Check the amino acid sequences you got from your code on the site below

Link: Amino acid sequence checker



Choose which sequence to check here

Contents

1 The Central Dogma (Flow of Genetic Information)

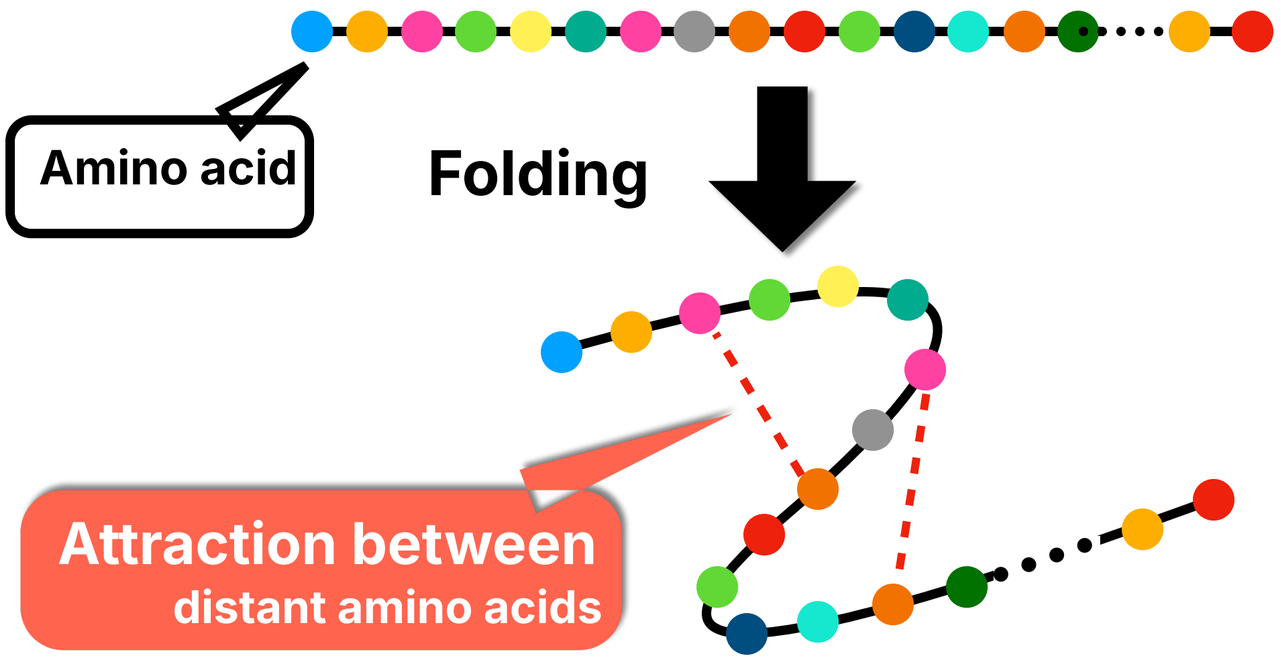
2 Reproducing the Central Dogma in Python

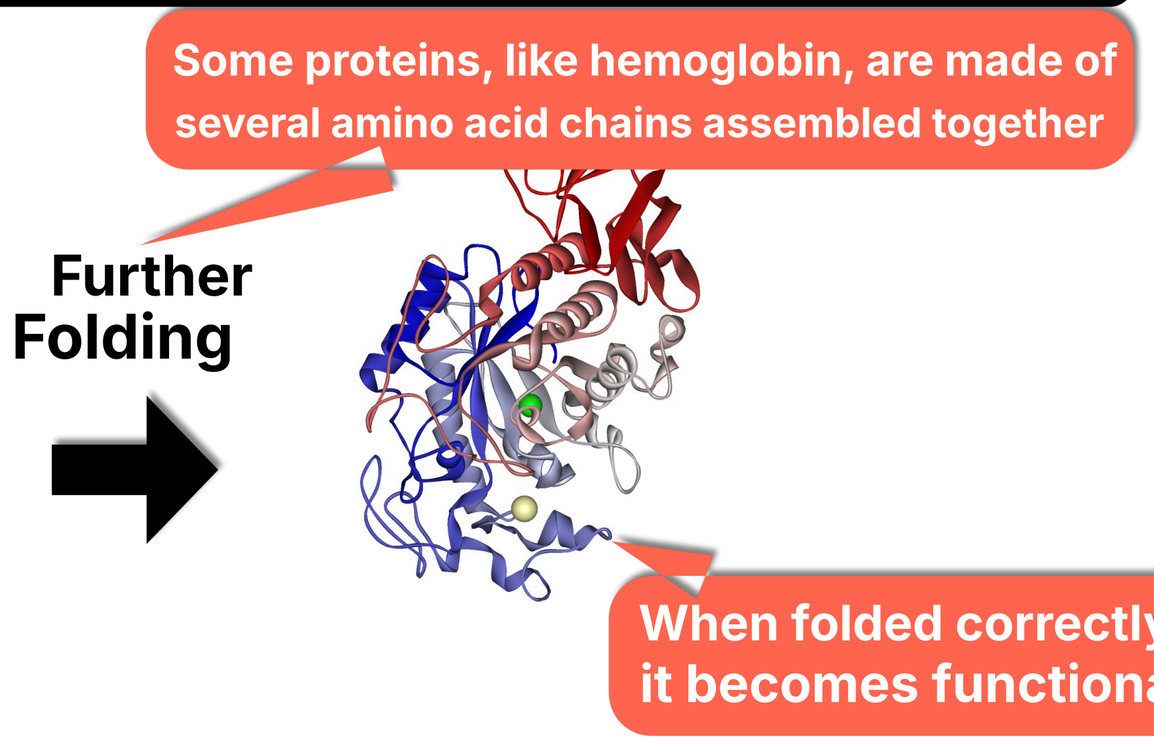
3 Why Protein Structure Matters, and AlphaFold

4 Applications in Medicine and Pharmacology

Protein

Amino acids: molecules (20 kinds) that link up in chains of **tens to thousands**
The type and order of amino acids determine how the chain folds and what it does



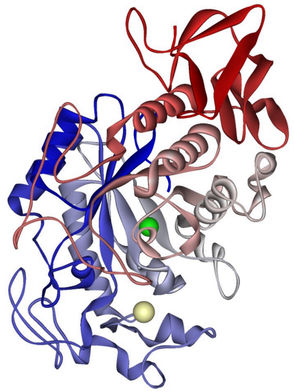


Some amino acids are attracted to distant amino acids in the chain
As a result, the amino acid chain folds into a functional protein

The Importance of Protein Structure

Proteins bind to other substances and do their jobs through the structures they fold into,
so if the structure is wrong, the function is lost or impaired

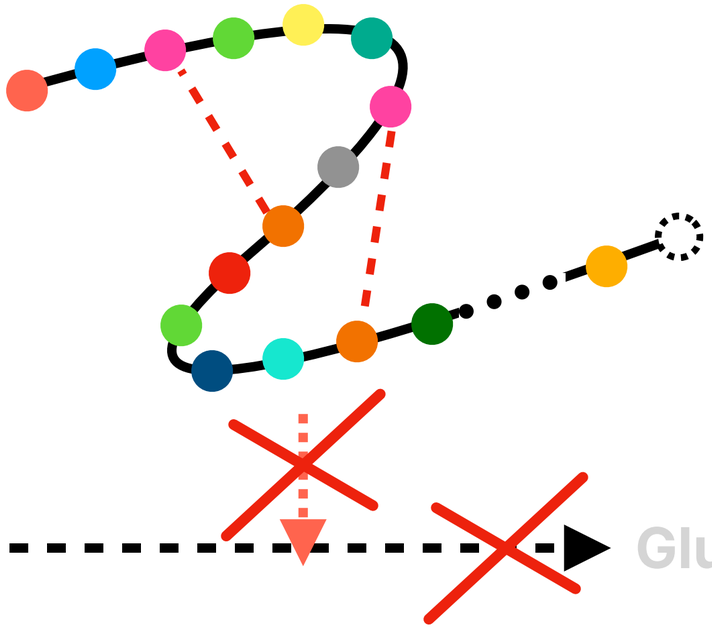
Amylase with the correct structure



Binding

Starch → Glucose

Amylase with a different structure



Starch - - - ✕ - - - ✕ Glucose

Biological significance: knowing the structure is essential to understanding a protein's function

Medical significance: changes in protein structure cause many diseases

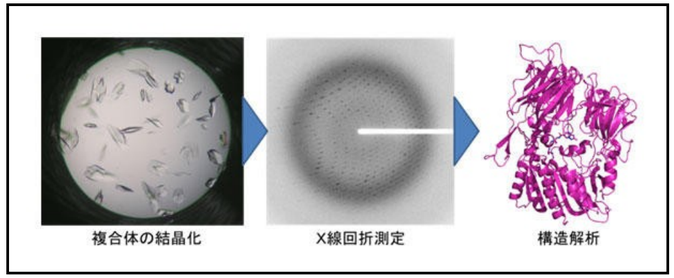
Methods for Protein Structure Analysis

Vital, yet technology lagged
A 50-year-old biology problem

Existing Methods for Analyzing Protein Structure

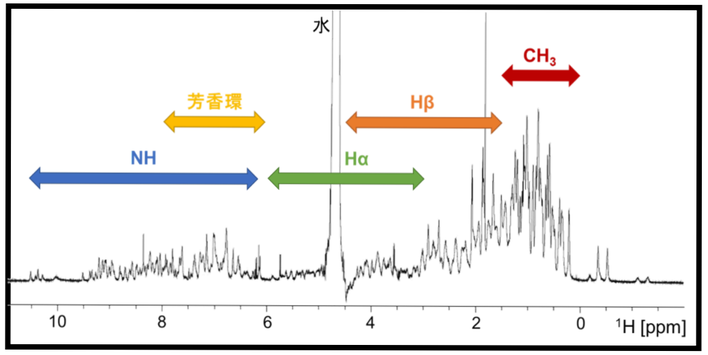
There are three main ways to determine a protein's 3D structure
However, every method has drawbacks, and structure analysis was costly

X-ray crystallography



The most common method, but
Analyzing a protein structure
takes **months to years**

NMR spectroscopy



Relatively low cost, but
only small proteins can be analyzed

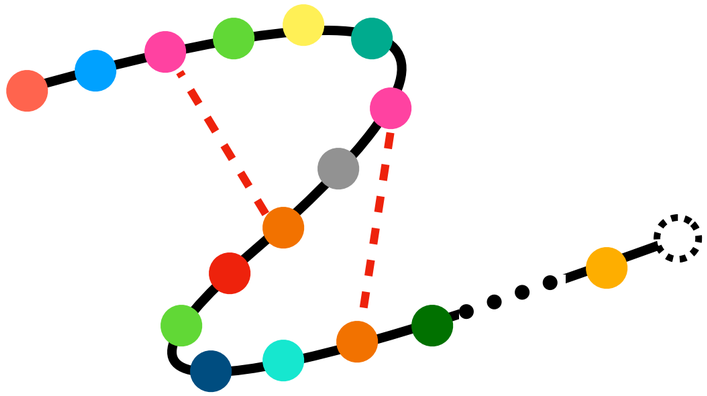
Cryo-electron microscopy



A Nobel Prize-winning technique
for high-resolution analysis
But the equipment is very costly

Protein Structure Prediction AI ~AlphaFold~

Trained on amino acid sequences and their 3D structures determined by existing methods,
AlphaFold was born as an AI that predicts protein structures with high accuracy

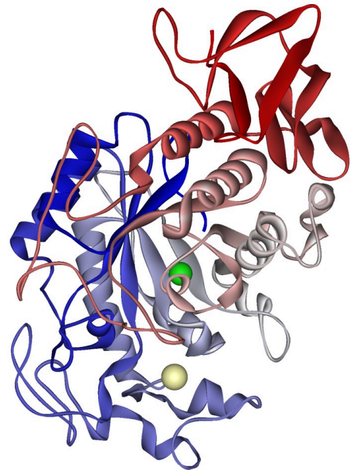


Amino acid sequence
(Example: MVHLTPV・・・)

Input

AlphaFold

Output



Protein structure

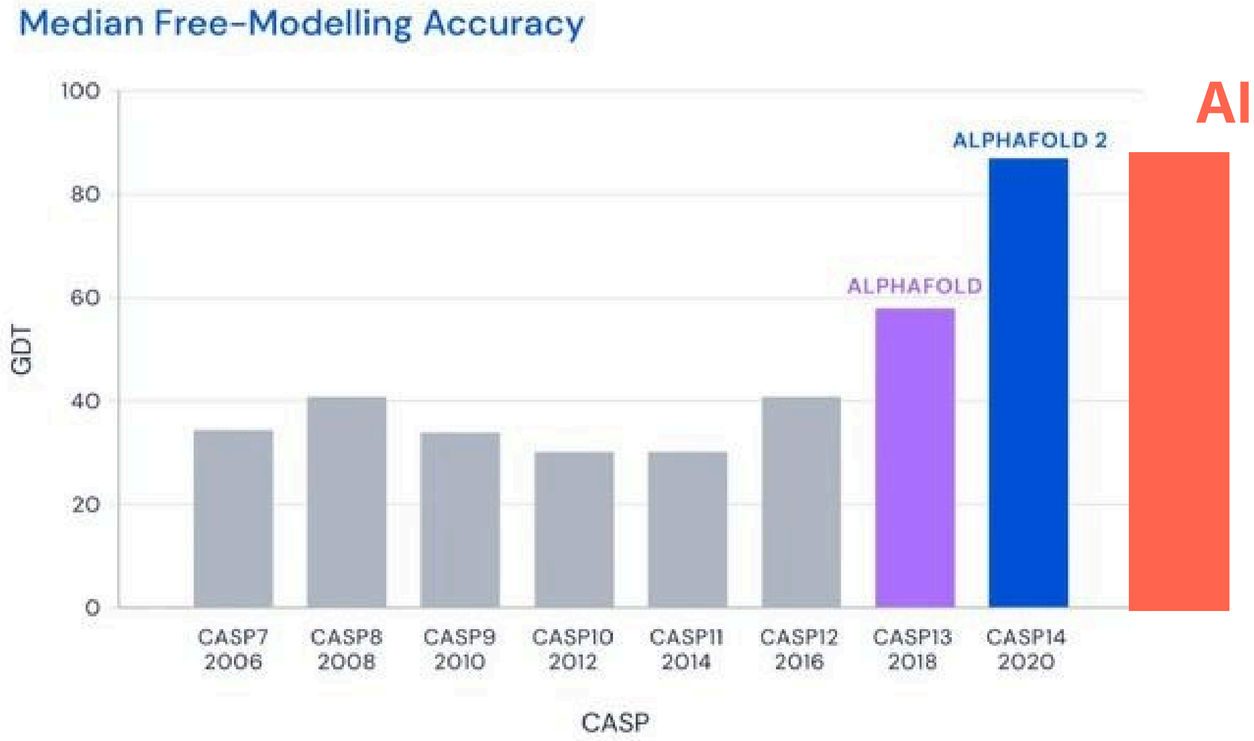
For helping to solve protein structure prediction, a 50-year-old problem in biology,
it won the 2024 Nobel Prize in Chemistry

AlphaFold's Prediction Accuracy

In CASP, the key competition for protein structure prediction accuracy, AlphaFold 2

predicted structures with an average similarity of 92.4 GDT to those obtained by experiment (the 3 methods above)

Similarity to real structure



AlphaFold 3

Better binding prediction

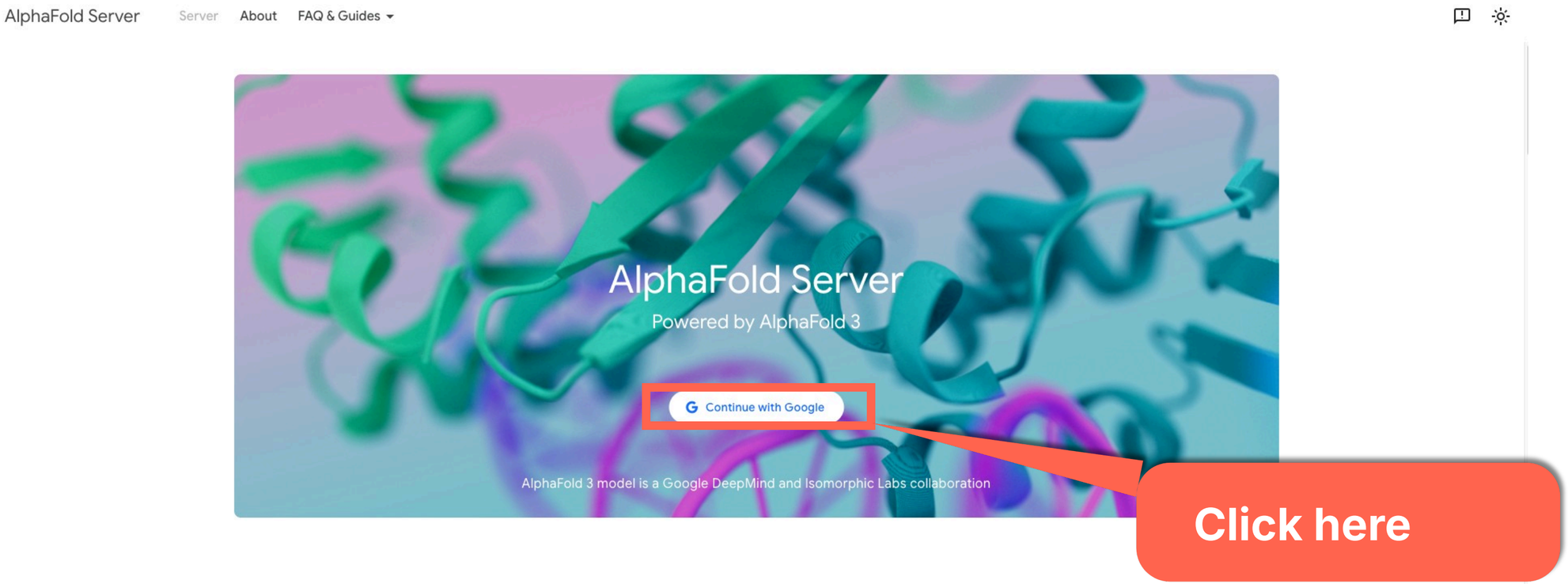
Latest model

Anyone can use AlphaFold!!

How to Use AlphaFold 3

To use the latest model, AlphaFold 3, open the AlphaFold Server page

AlphaFold Server



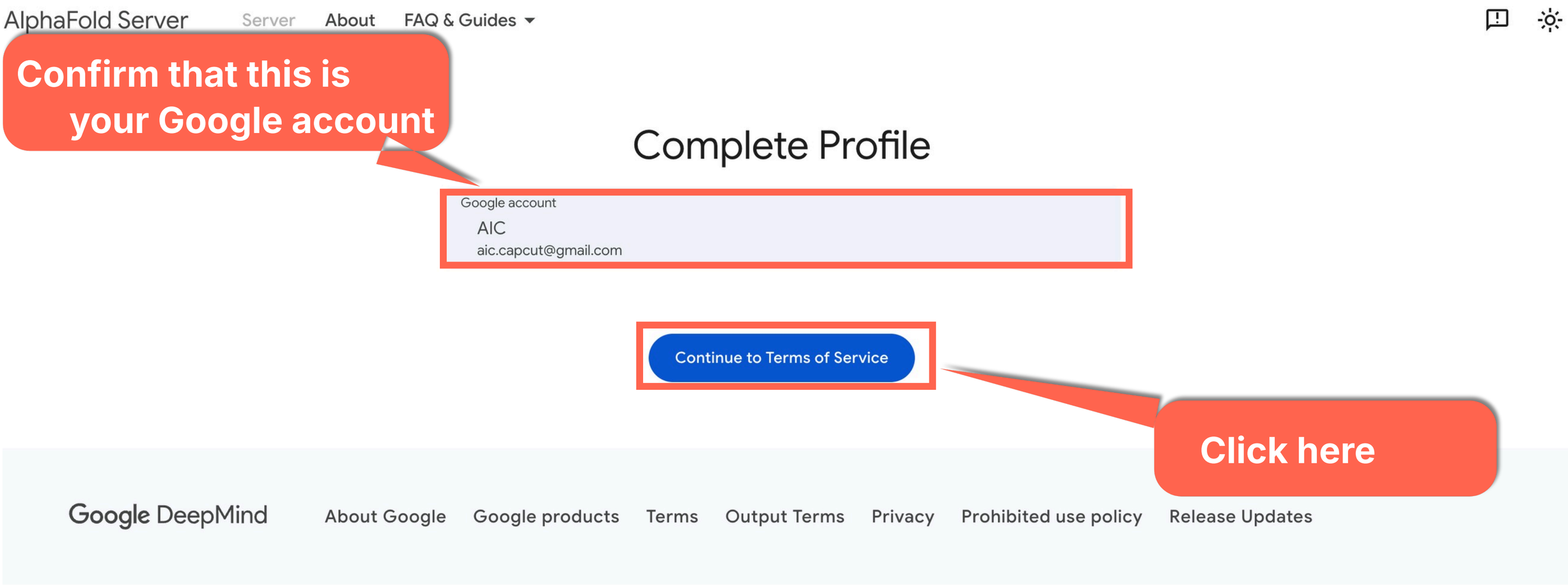
How to Use AlphaFold 3

If you are using a loaned AIC PC, add and use your own Google account



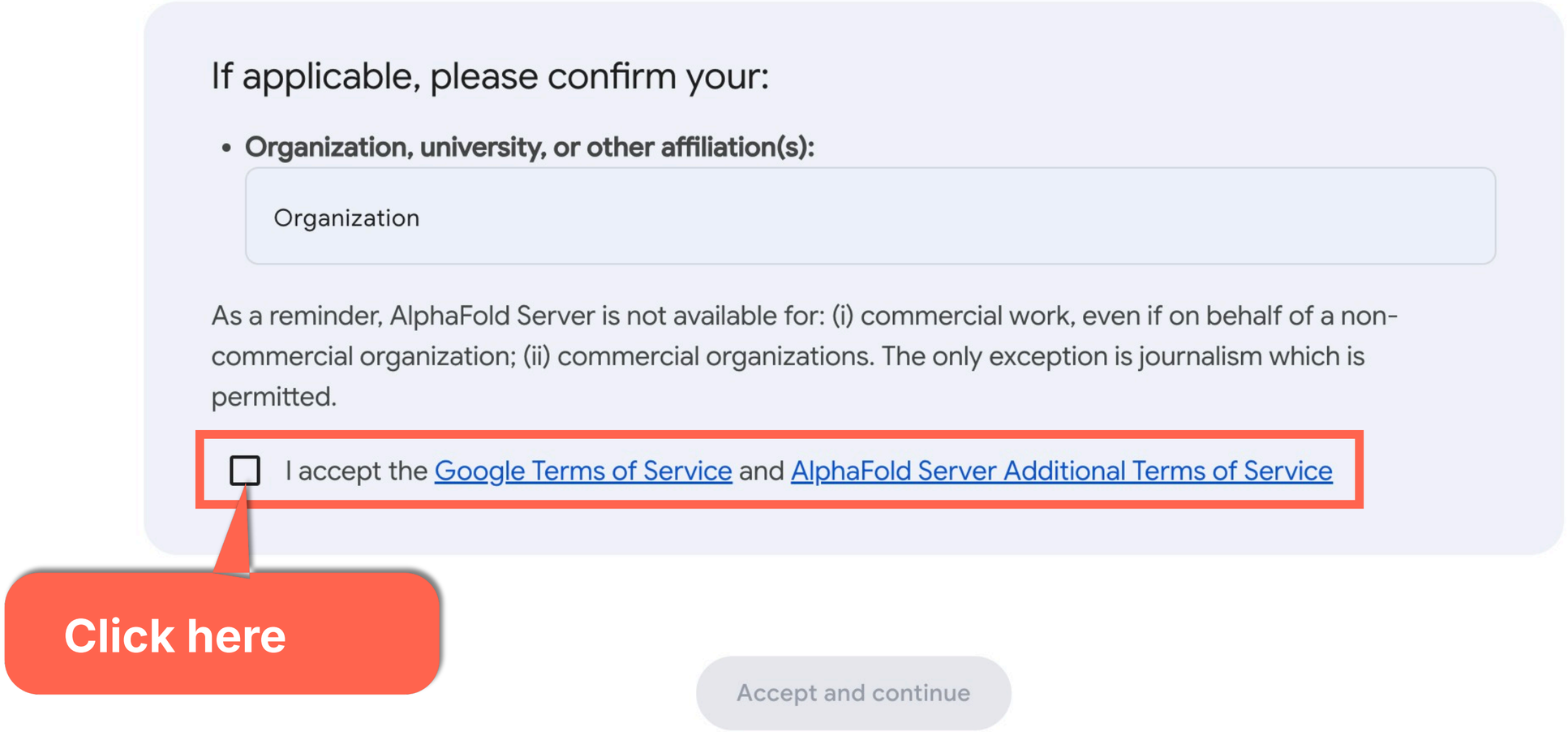
How to Use AlphaFold 3

Confirm that your Google account is shown, then click the blue button



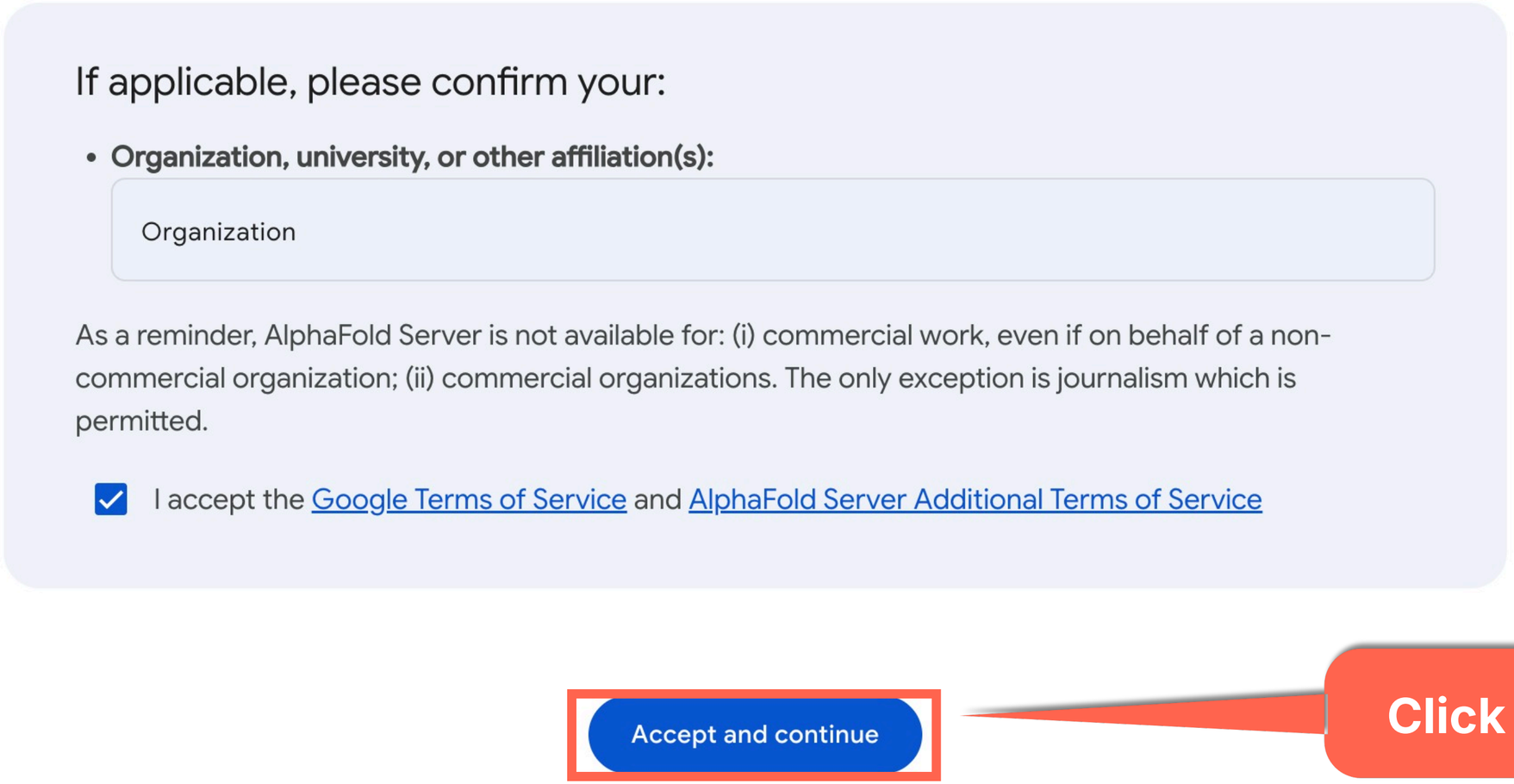
How to Use AlphaFold 3

On the next page, tick the checkbox near the bottom



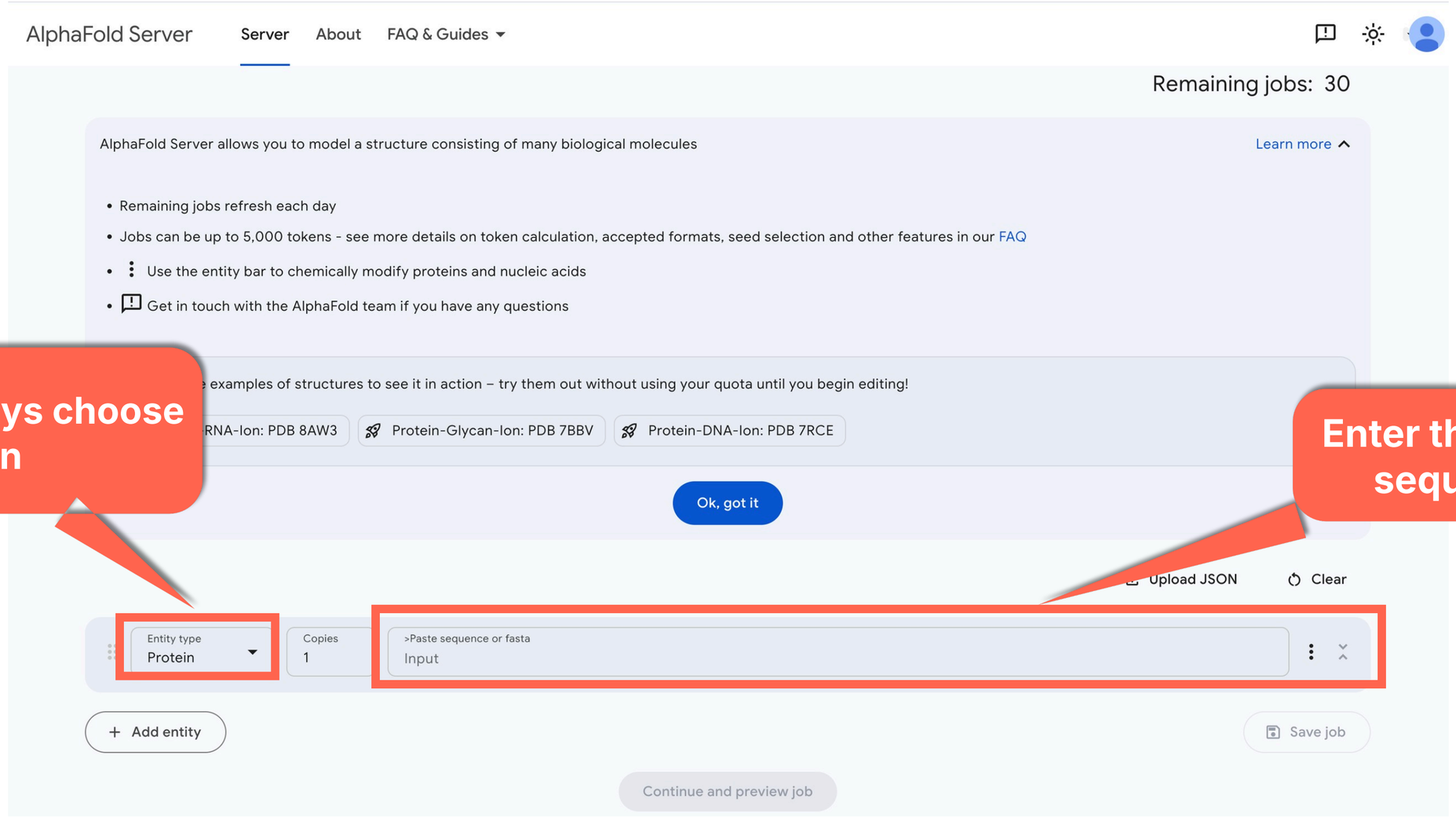
How to Use AlphaFold 3

On the next page, tick the checkbox near the bottom



How to Use AlphaFold 3

Once logged in, you will see a screen like the one below; enter sequences in the box at the bottom



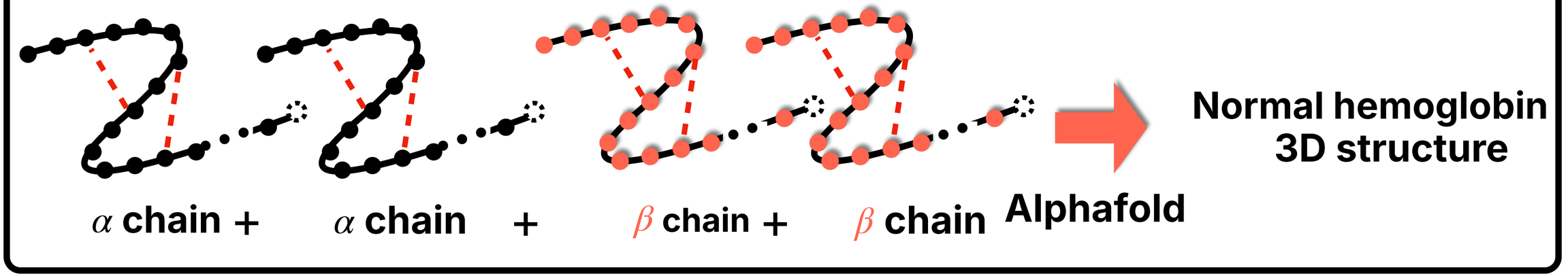
This time, always choose Protein

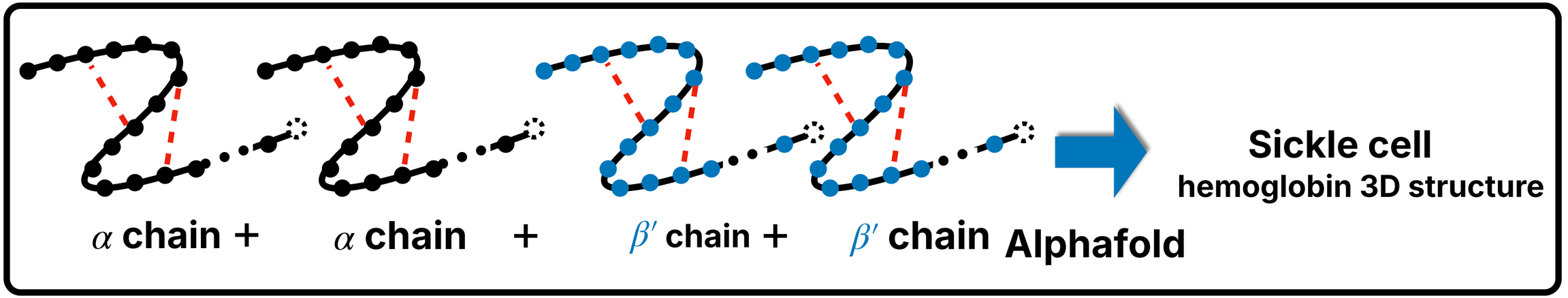
Enter the amino acid sequence here

AlphaFold Exercise

Using the 3 amino acid sequences created with your code in the Python exercise,

let's predict the structures of normal hemoglobin and hemoglobin from sickle cell disease patients





Predicting the Structure of Normal Hemoglobin

Enter the α chain and β chain sequences as shown below

Two α chains

Copy α chain sequence

Entity type: Protein
Copies: 2

```
MVLSPADKTN VKAAWGKVGA HAGEYGAEAL ERMFLSFPTT KTYFPHFDLS HGSAQVKGHG
KKVADALTNA VAHVDDMPNA LSALSDLHAH KLRVDPVNFK LLSHCLLVTL AAHLPAEFTP
AVHASLDKFL ASVSTVLTSK YR
```

Entity type: Protein
Copies: 2

```
MVHLTPEEKS AVTALWGKVN VDEVGGEALG RLLVVYPWTQ RFFESFGDLS TPDAVMGNPK
VKAHGKKVLG AFSDGLAHLD NLKGTFATLS ELHCDKLHVD PENFRLLGNV LVCVLAHHFG
KEFTPPVQAA YQKVVAGVAN ALAHKYH
```

Two β chains

Copy β chain sequence

+ Add entity

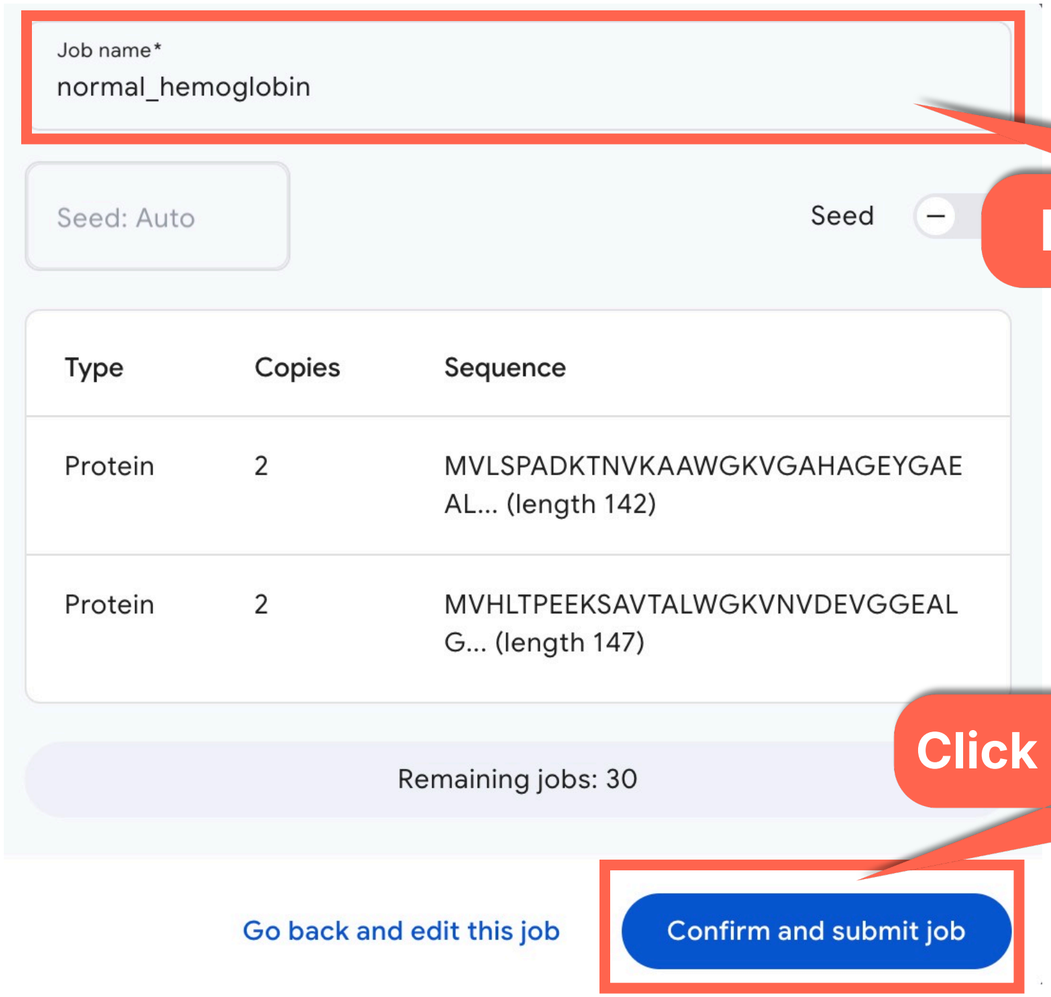
Save job

Continue and preview job

Click here

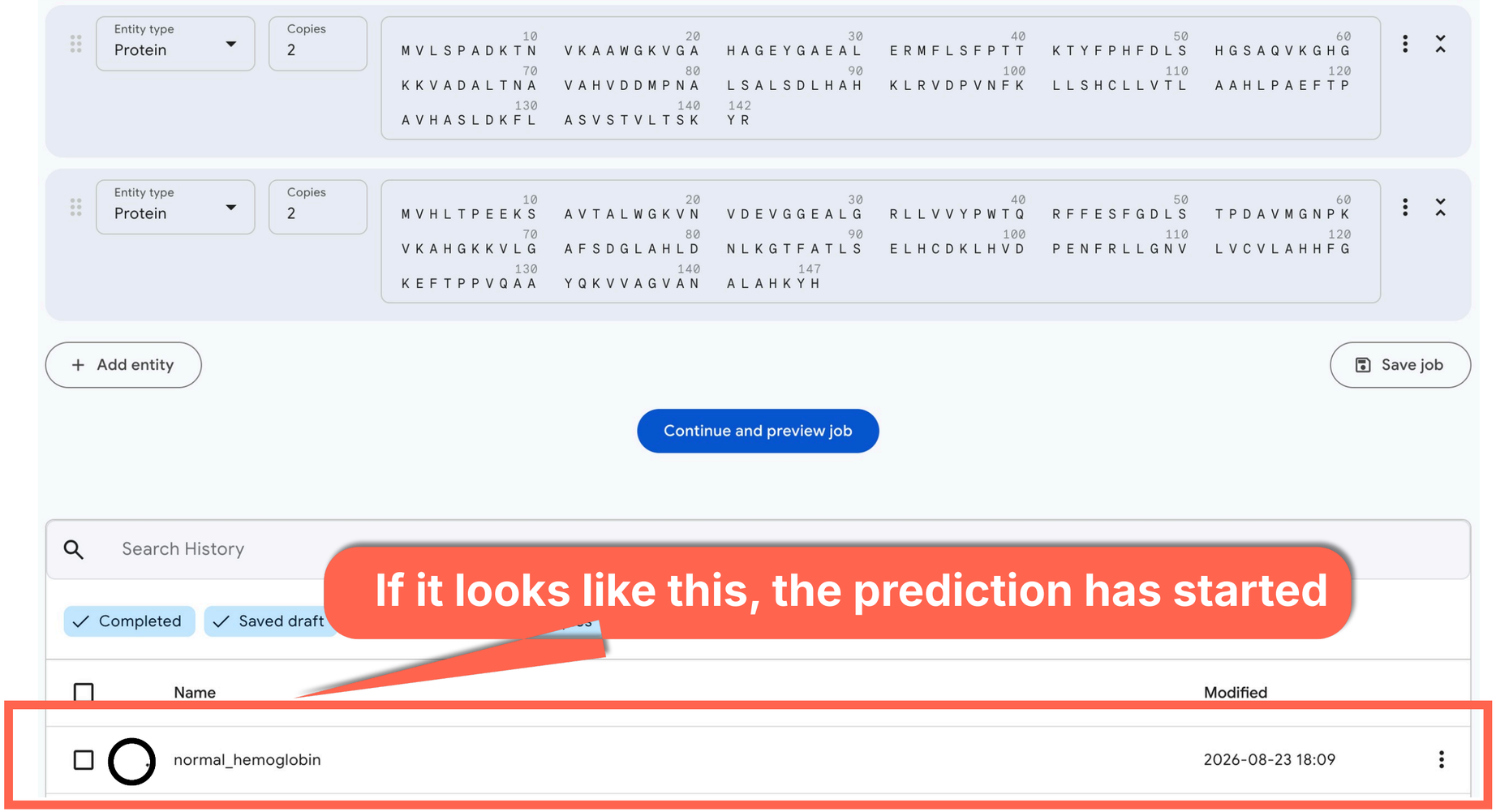
Predicting the Structure of Normal Hemoglobin

Enter the α chain and β chain sequences as shown below



Predicting the Structure of Normal Hemoglobin

Enter the α chain and β chain sequences as shown below



Predicting the Structure of Sickle Cell Hemoglobin

Enter the α chain and β' chain sequences as shown below

Two α chains

Copy α chain sequence

Entity type: Protein
Copies: 2

```
                10                  20                  30                  40                  50                  60
MVLSPADKTN  VKAAWGKVGA  HAGEYGAEAL  ERMFLSFPTT  KTYFPHFDLS  HGSAQVKGHG
                70                  80                  90                 100                 110                 120
KKVADALTNA  VAHVDDMPNA  LSALSDLHAH  KLRVDPVNFK  LLSHCLLVTL  AAHLPAEFTP
               130                 140                 142
AVHASLDKFL  ASVSTVLTSK  YR
```

Entity type: Protein
Copies: 2

```
                10                  20                  30                  40                  50                  60
MVHLTPVEKS  AVTALWGKVN  VDEVGGEALG  RLLVVYPWTQ  RFFESFGDLS  TPDAVMGNPK
                70                  80                  90                 100                 110                 120
VKAHGKKVLG  AFSDGLAHLD  NLKGTFATLS  ELHCDKLHVD  PENFRLLGNV  LVCVLAHHFG
               130                 140                 147
KEFTPPVQAA  YQKVVAGVAN  ALAHKYH
```

Two β chains

Just swap in the β' chain

+ Add entity

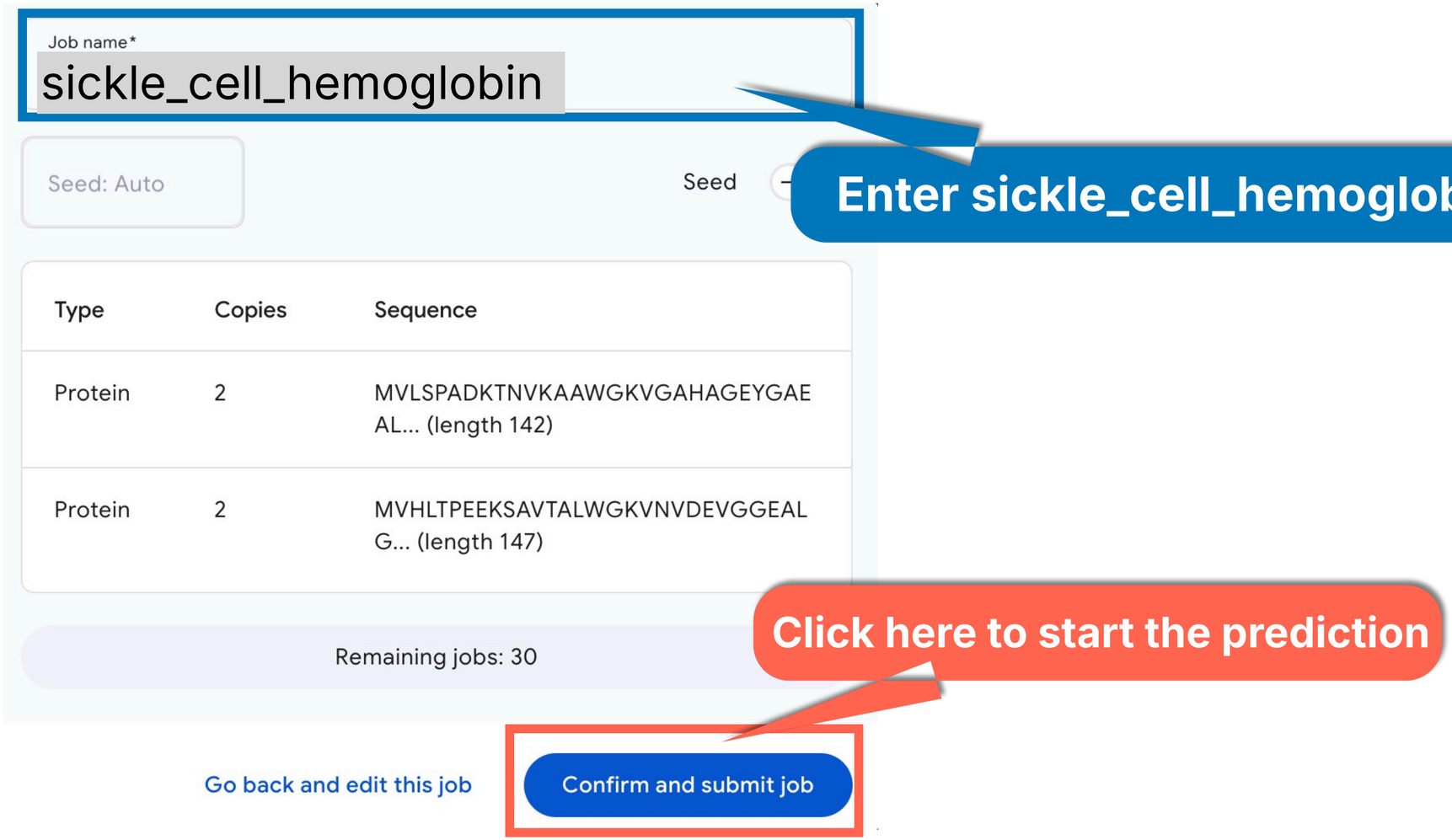
...ve job

Continue and preview job

Click here

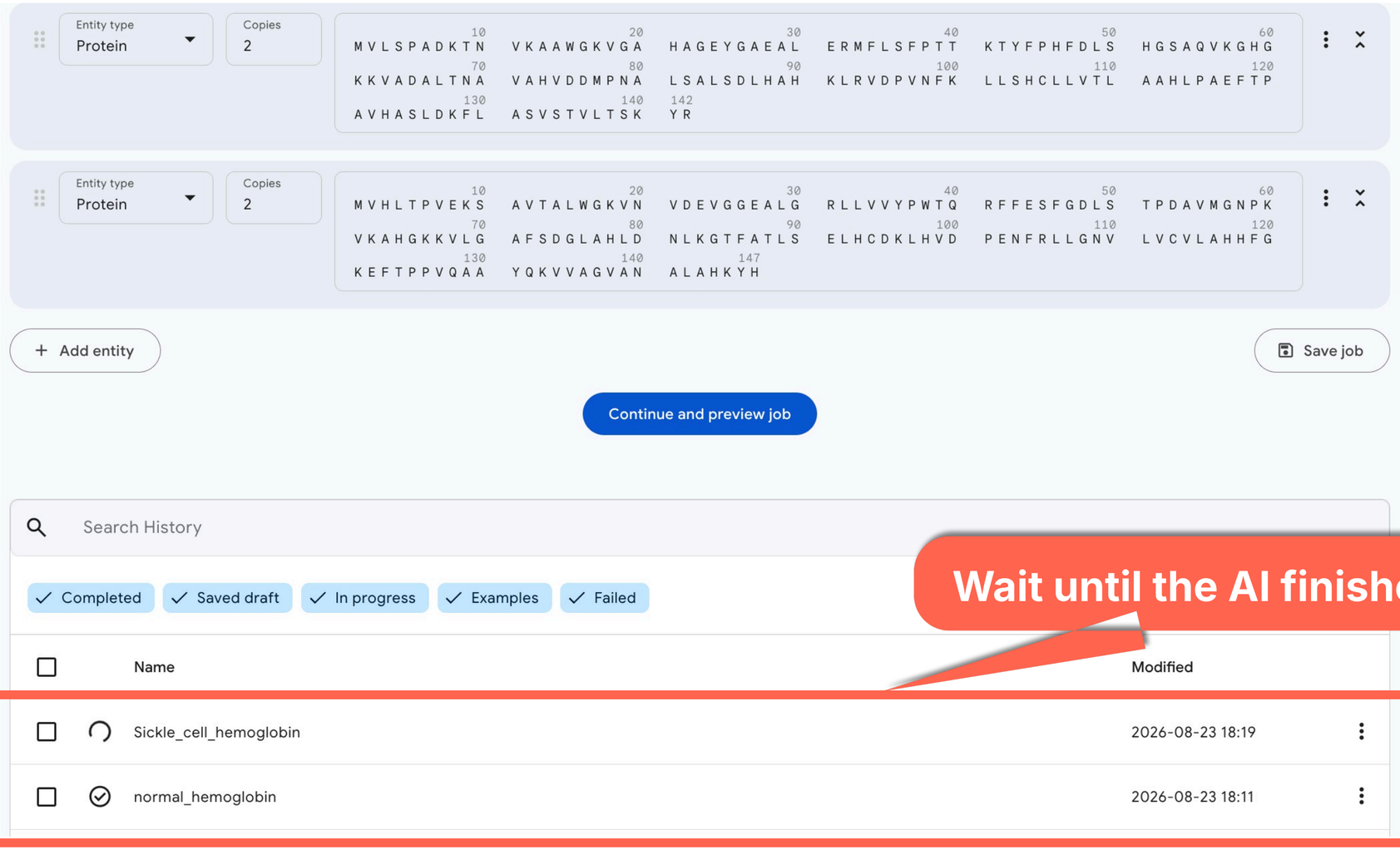
Predicting the Structure of Sickle Cell Hemoglobin

Enter the α chain and β' chain sequences as shown below



3D Structure Prediction

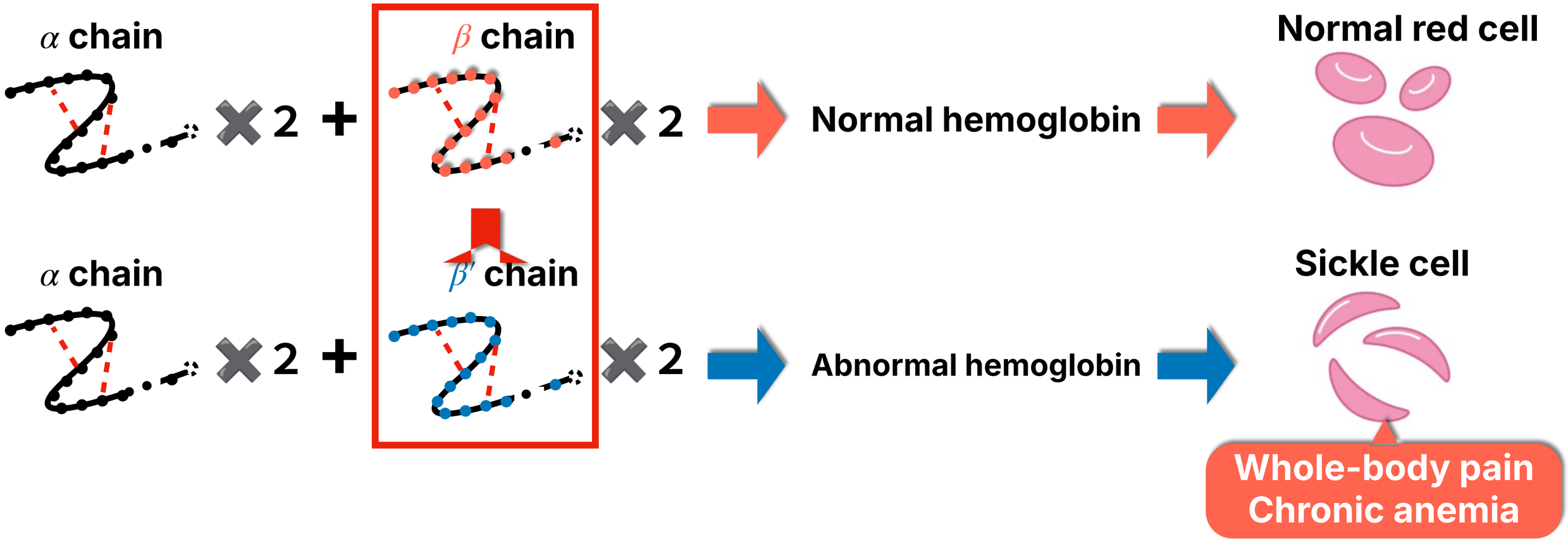
Now we have started predicting the two structures:
normal hemoglobin and sickle cell hemoglobin



Wait until the AI finishes the prediction!

Sickle Cell Disease

A DNA change (mutation) in the hemoglobin β chain yields an abnormal chain, β',
an inherited disease caused by a change in hemoglobin's structure



Mutation

The hemoglobin β chain becomes β' through a point mutation: a one-letter DNA change

Original DNA sequence

...TGACTCCTG**A**GGAGAAG...
92

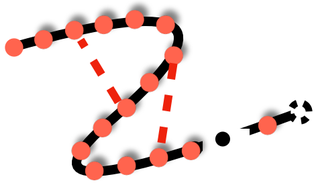
Point mutation

Patient's DNA sequence

...TGACTCCTG**T**GGAGAAG...
92

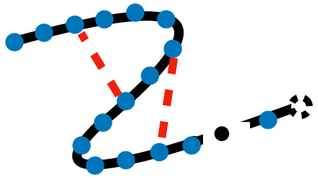
Transcription & translation

MVHLTP**E**EKSAVTAL...
7



β chain

MVHLTP**V**EKSAVTAL...
7

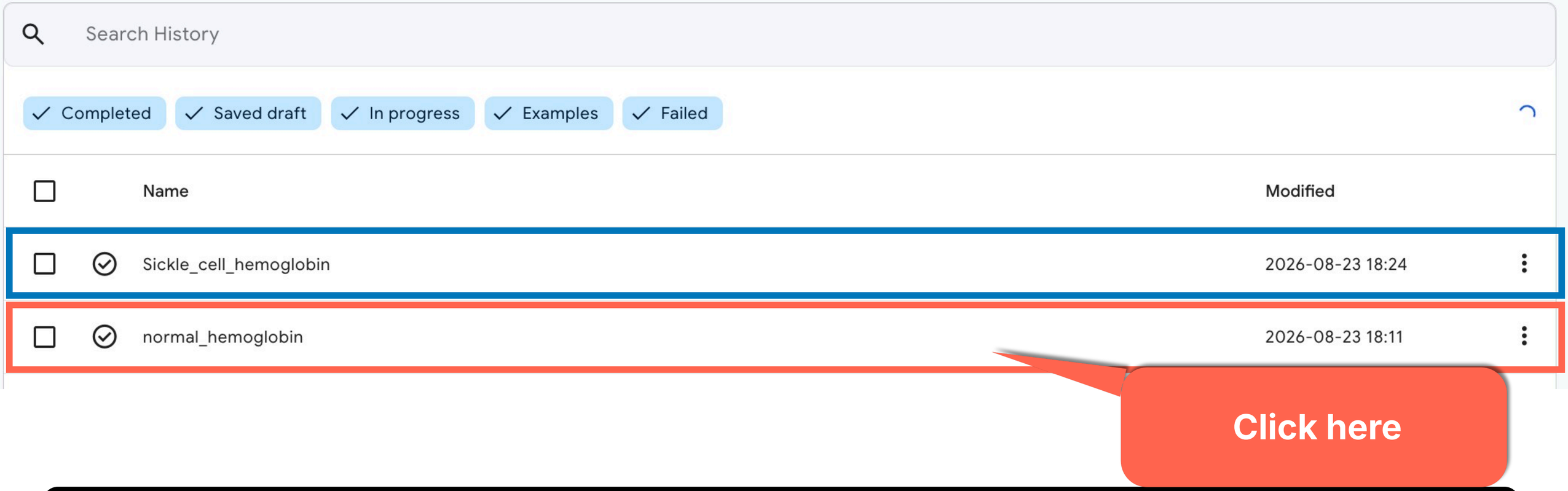


β' chain

Why does changing just one letter of DNA cause sickle cell disease? The key lies in the 3D structure

Structure Prediction Results

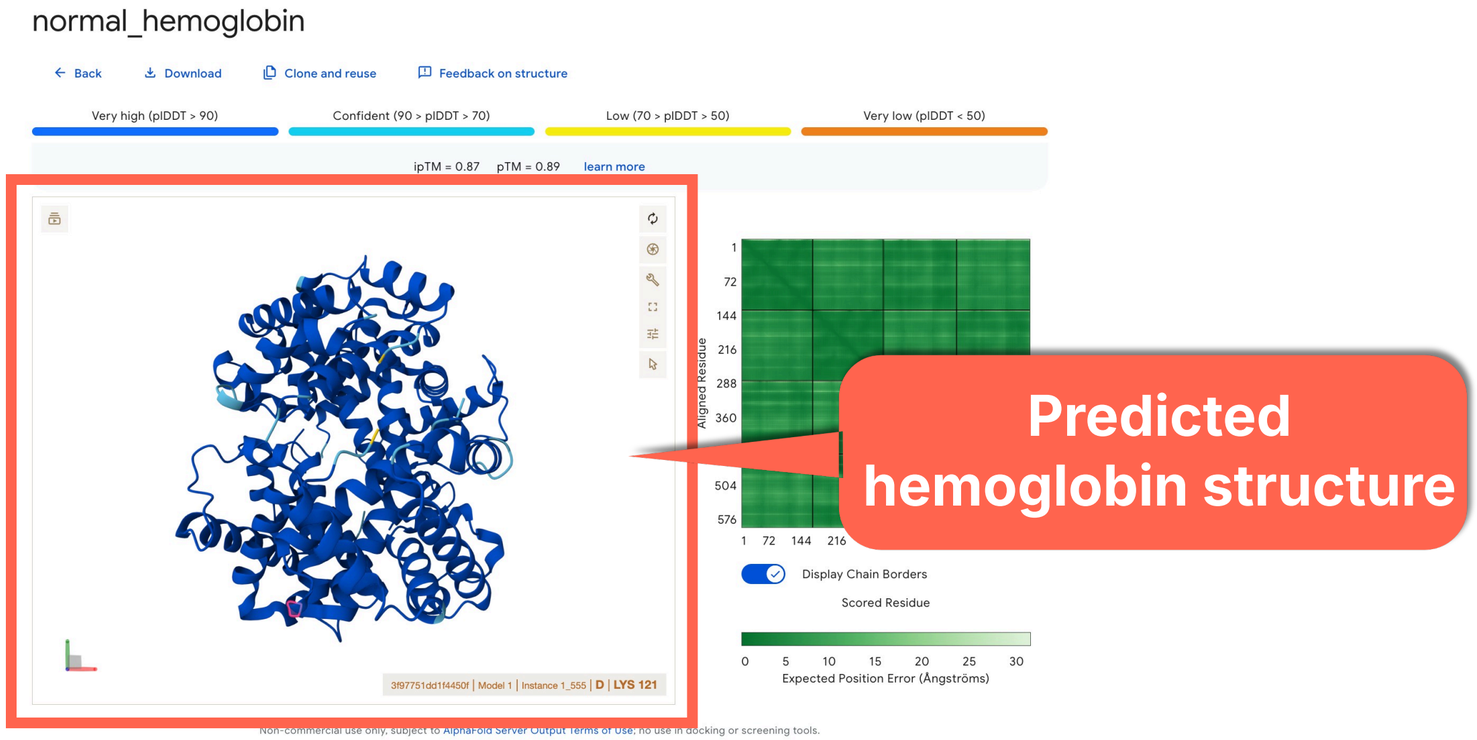
When the run finishes, click it to check the prediction results



How can we compare the predicted structures of normal hemoglobin and sickle cell hemoglobin?

Structure Prediction Results

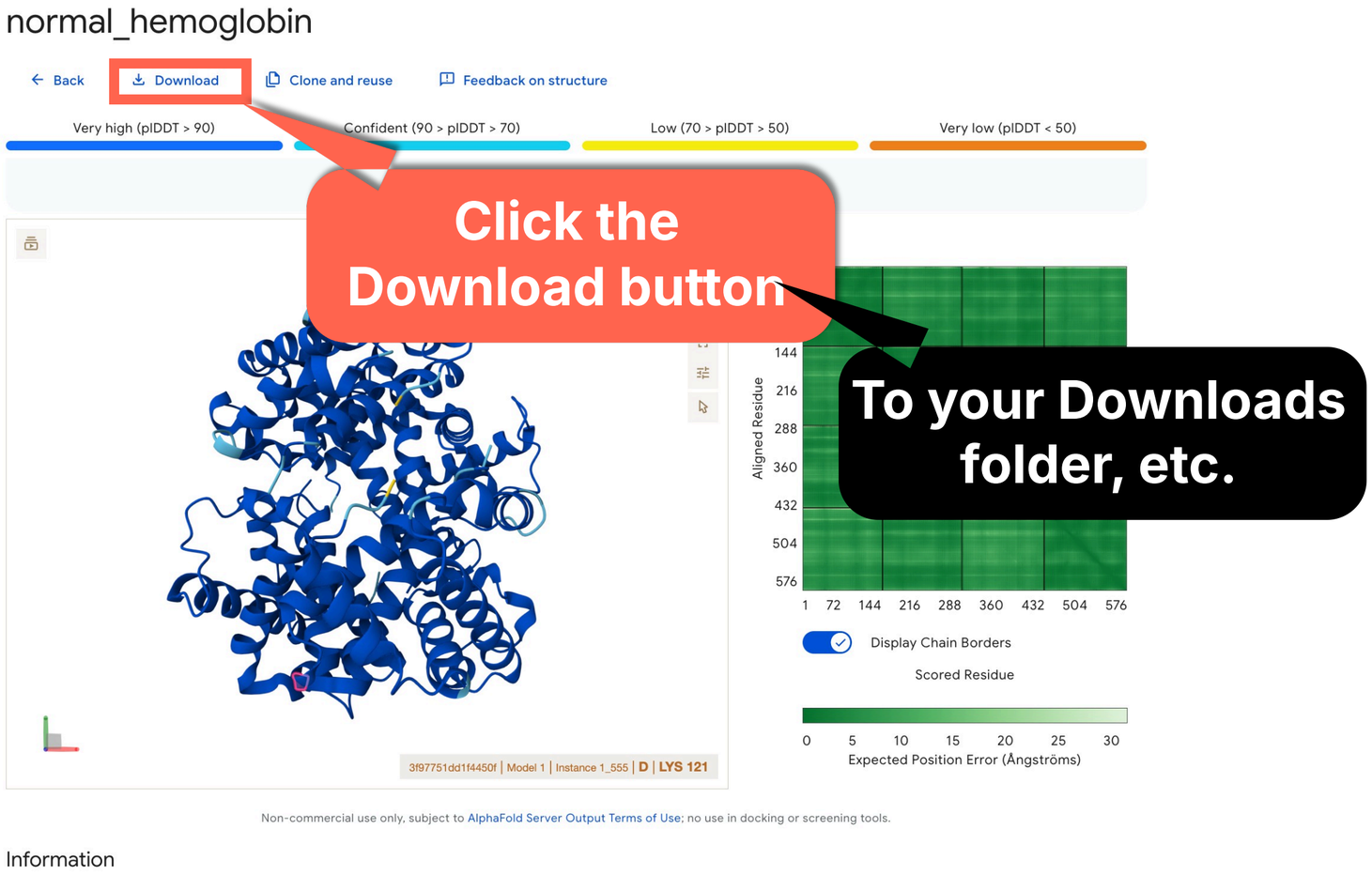
When the run finishes, click it to check the prediction results



How can we compare the predicted structures of normal hemoglobin and sickle cell hemoglobin?

Downloading the Results

Click the Download button near the top of the page
After a short wait, a zip file is downloaded to your PC

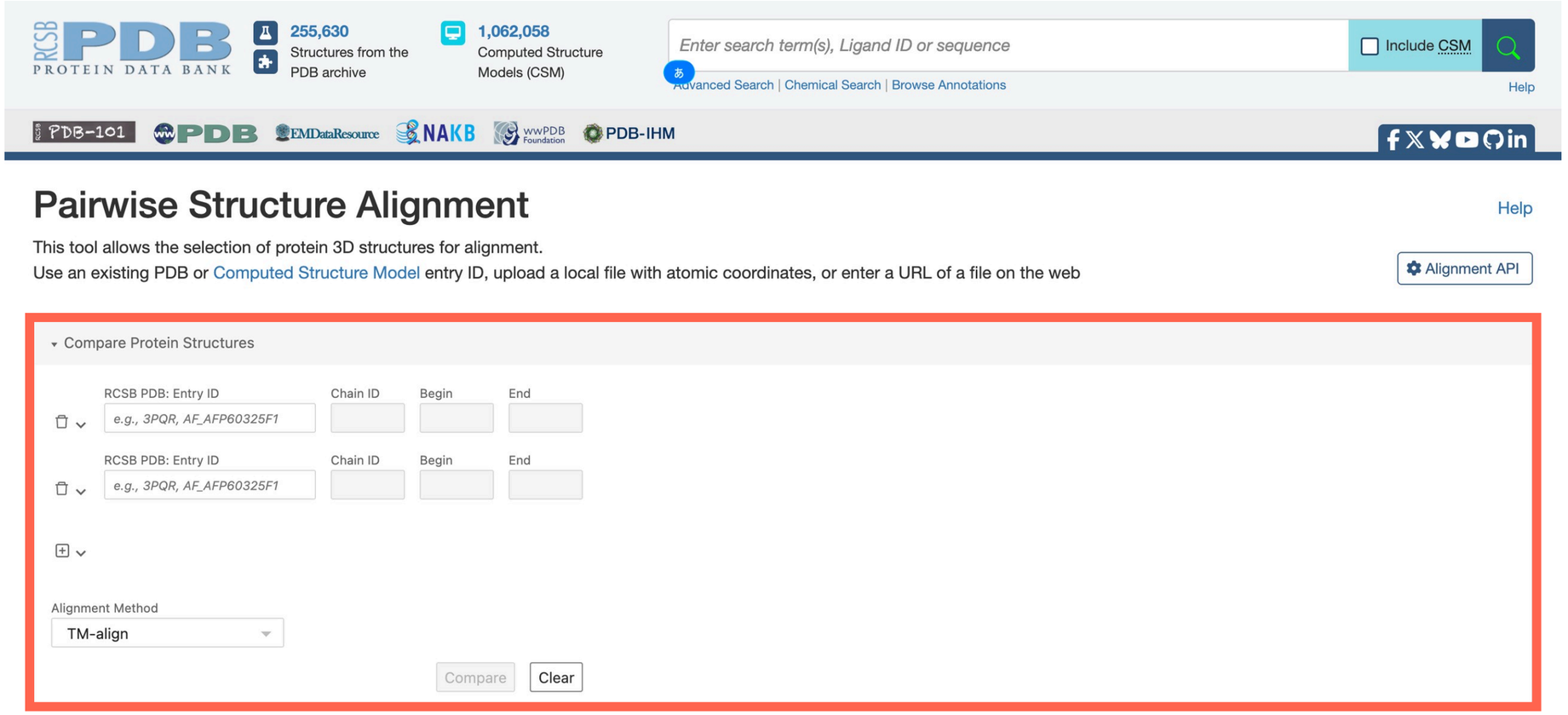


Structure Comparison

Overlay the structures of normal hemoglobin and sickle cell hemoglobin

Run a structure alignment with the tool below to compare the structures

RCSB PDB Pairwise Structure Alignment

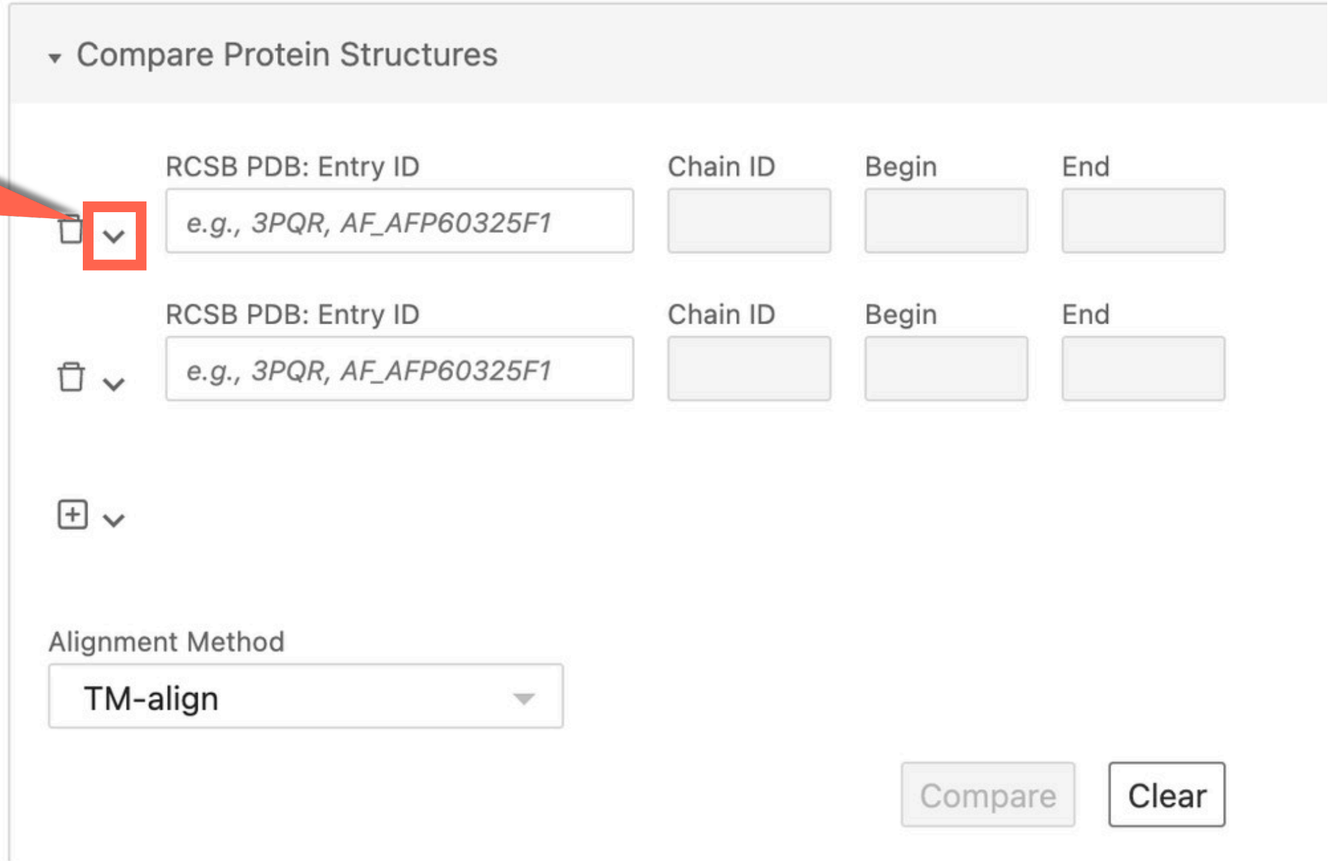


Using the Structure Comparison Tool

How to use

Enter the predicted structures (.pdb files) of normal and sickle cell hemoglobin in the fields below and run the structure alignment

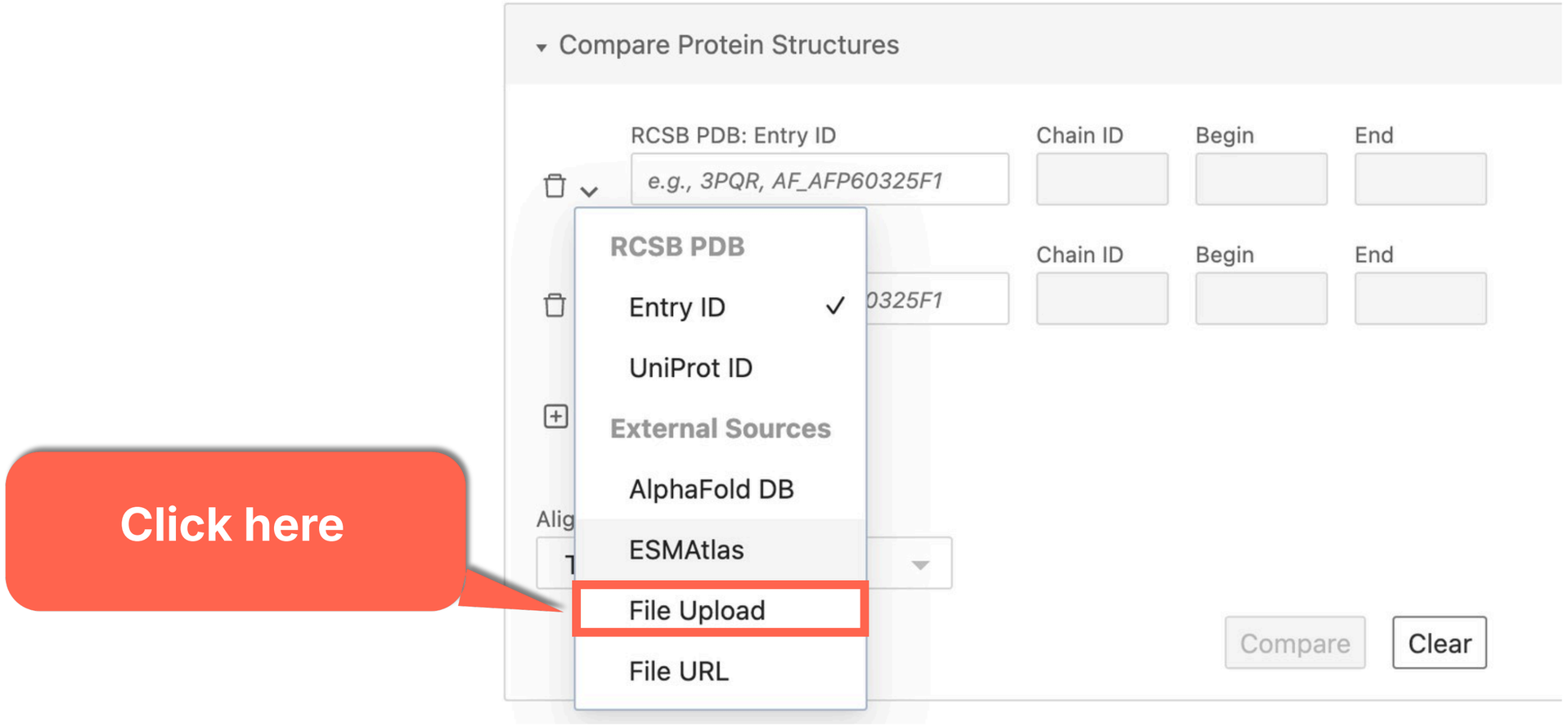
Click here



Using the Structure Comparison Tool

How to use

Enter the predicted structures (.pdb files) of normal and sickle cell hemoglobin in the fields below and run the structure alignment

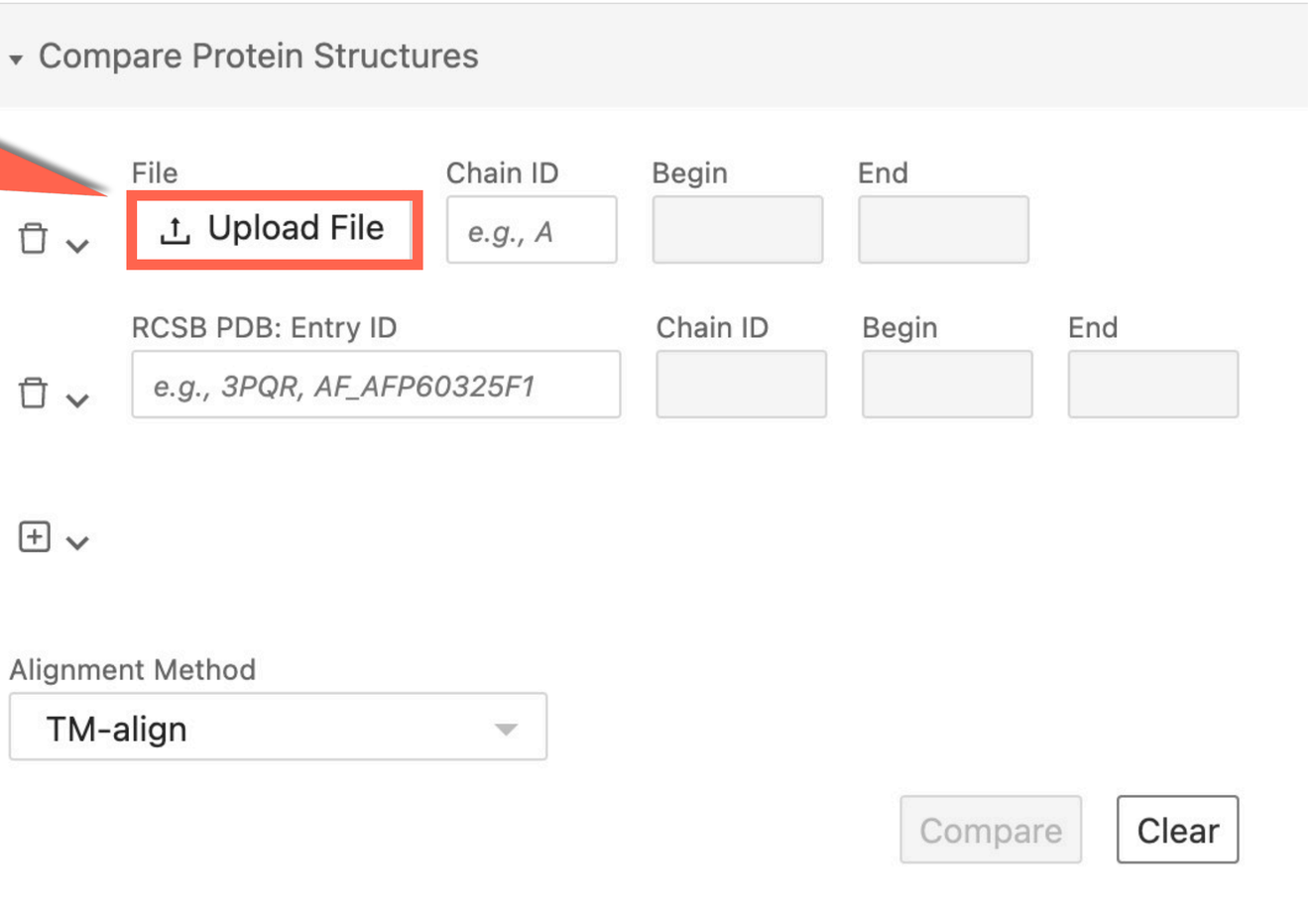


Using the Structure Comparison Tool

How to use

Enter the predicted structures (.cif files) of normal and sickle cell hemoglobin in the fields below and run the structure alignment

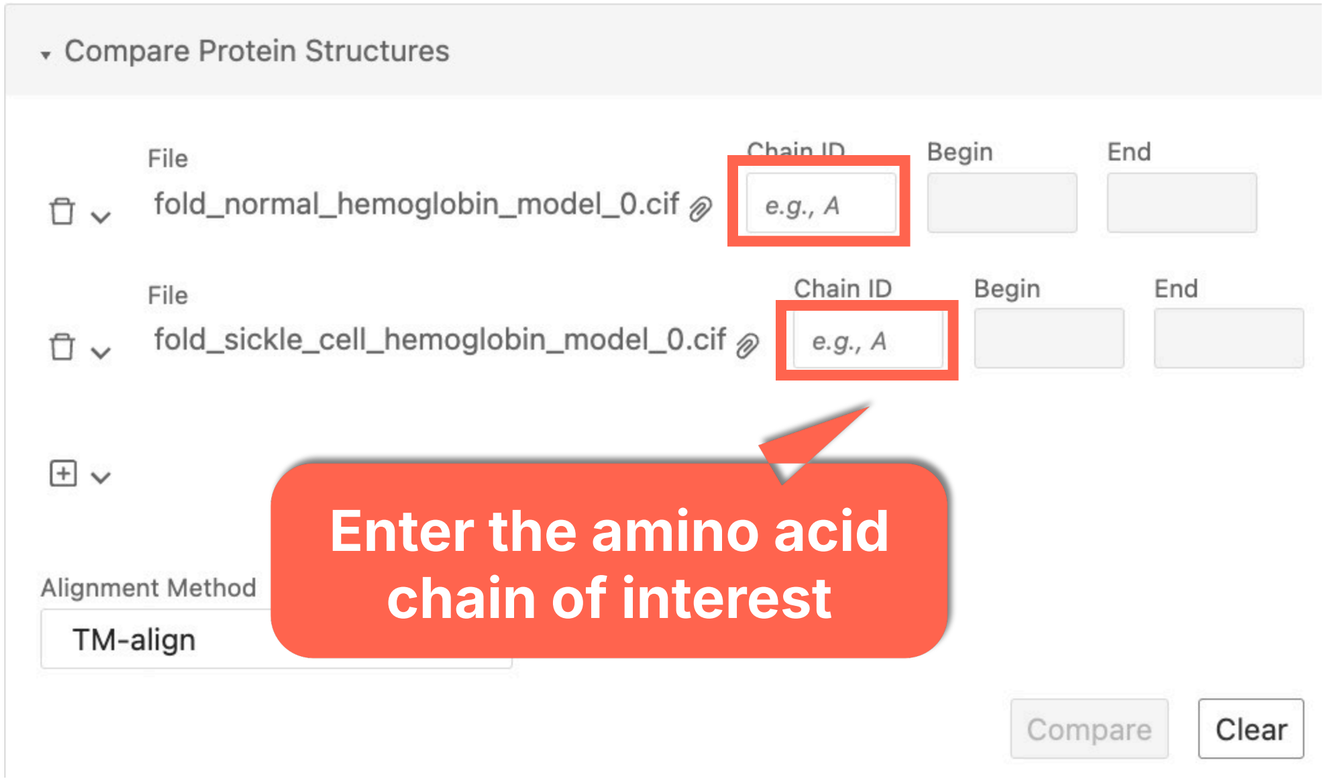
Select the .cif file



Using the Structure Comparison Tool

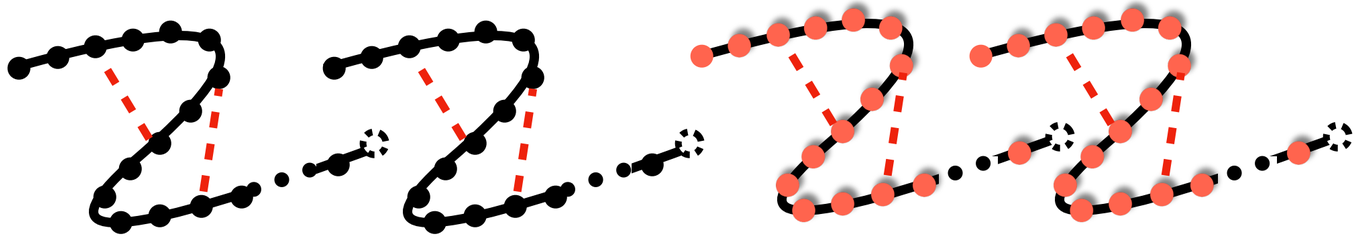
How to use

Enter the predicted structures (.pdb files) of normal and sickle cell hemoglobin in the fields below and run the structure alignment



The Amino Acid Chain of Interest

This time we want to focus on the β chain, which differs between the two kinds of hemoglobin

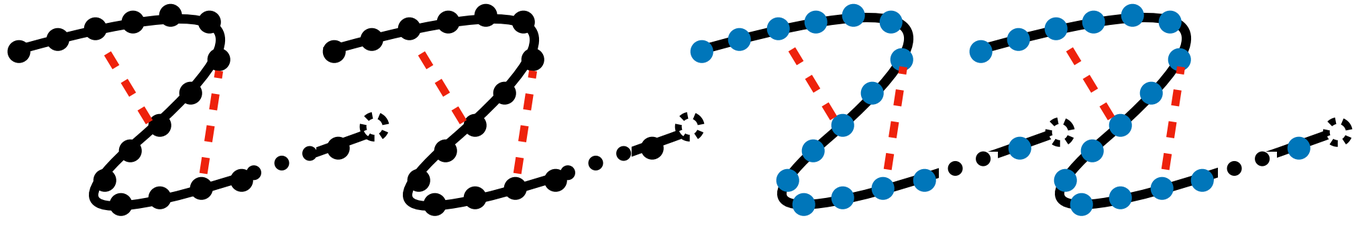


α chain + α chain + β chain + β chain

chain Id: A B C D

Alphafold

Normal hemoglobin 3D structure



α chain + α chain + β' chain + β' chain

chain Id: A B C D

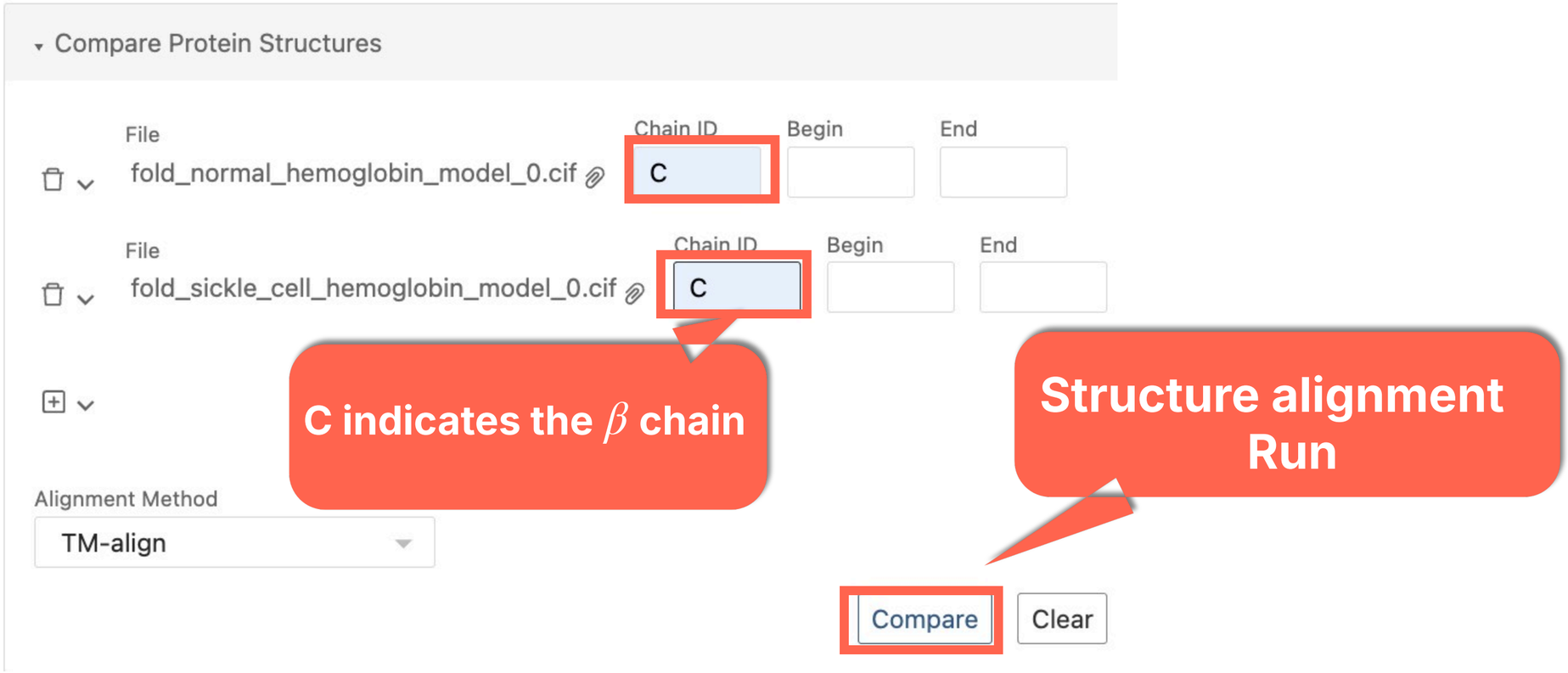
Alphafold

Sickle cell patient's hemoglobin 3D structure

Using the Structure Comparison Tool

How to use

Enter the predicted structures (.pdb files) of normal and sickle cell hemoglobin in the fields below and run the structure alignment



Reading the Results

The structure alignment results look like this

Pairwise Structure Alignment

Help

This tool allows the selection of protein 3D structures for alignment.
Use an existing PDB or Computed Structure Model entry ID, upload a local file with atomic coordinates, or enter a URL of a file on the web

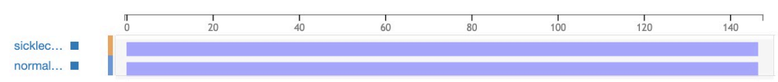
Alignment API

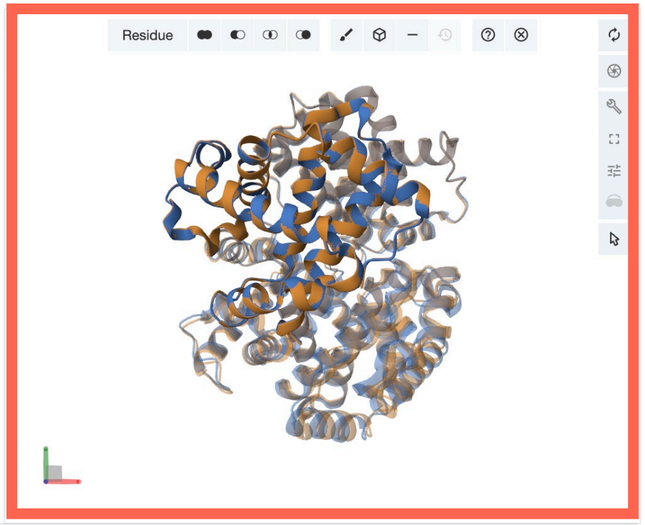
▸ Compare Protein Structures

Entry	Chain	RMSD	TM-score	Identity	Aligned Residues	Sequence Length	Modeled Residues
sicklecellhemoglobin_d4d1e_unrelaxed_rank_001_alphafold2_multimer_v3_model_2_seed_000.pdb	C	-	-	-	-	147	147
normalhemoglobin_608df_unrelaxed_rank_001_alphafold2_multimer_v3_model_3_seed_000.pdb	C	0.14	1	99%	147	147	147

Sequence Alignment in 3D

Export Copy Link





Overlaid structures

Zoom-In Feature

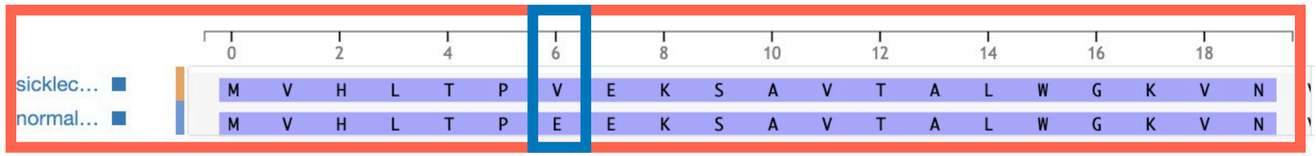
A feature that focuses on a specific amino acid in the amino acid chain of interest



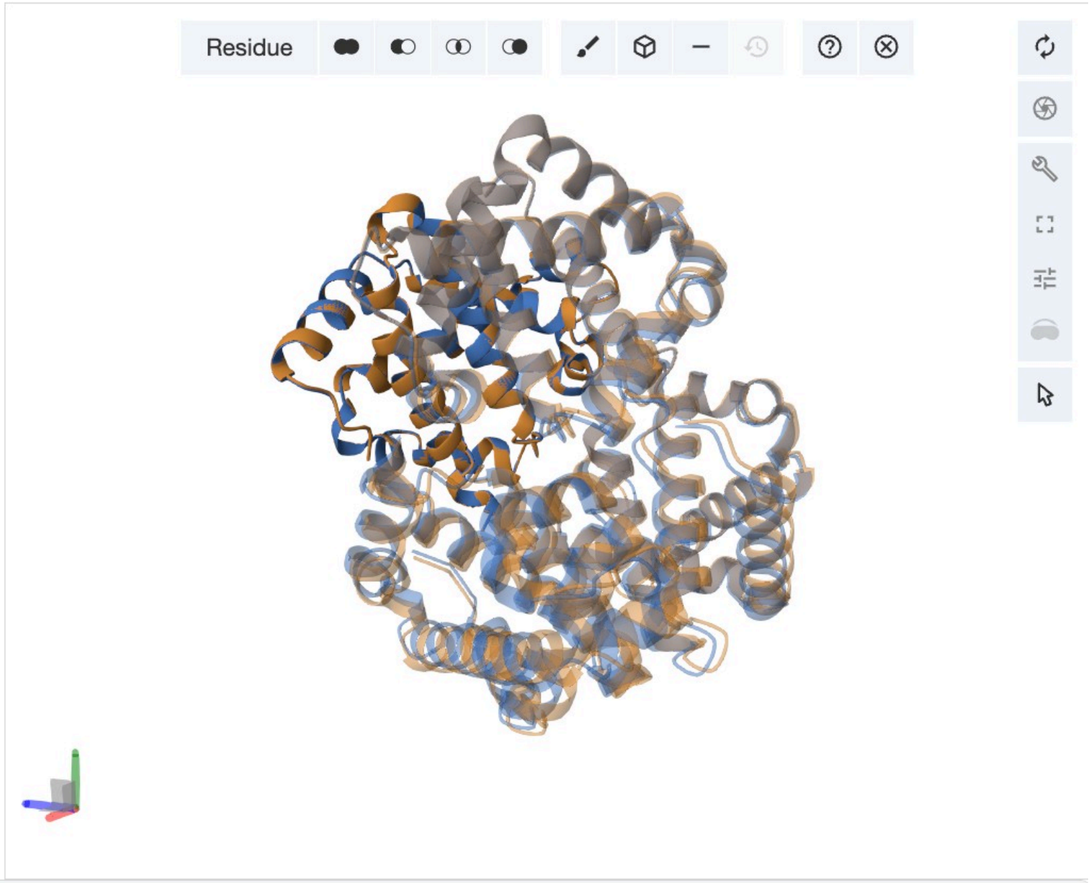
Zoom-In Feature

A feature that focuses on a specific amino acid in the amino acid chain of interest

Sequence Alignment in 3D

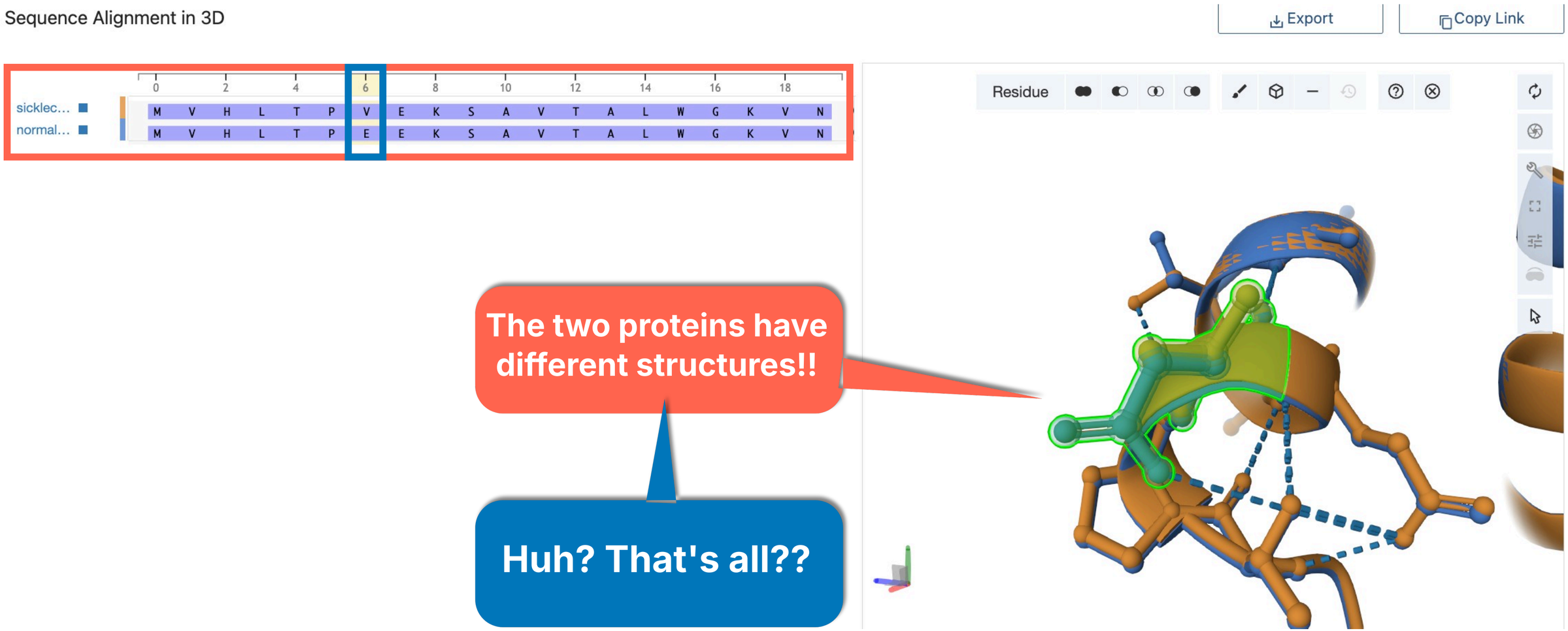


Click the part with the mutation



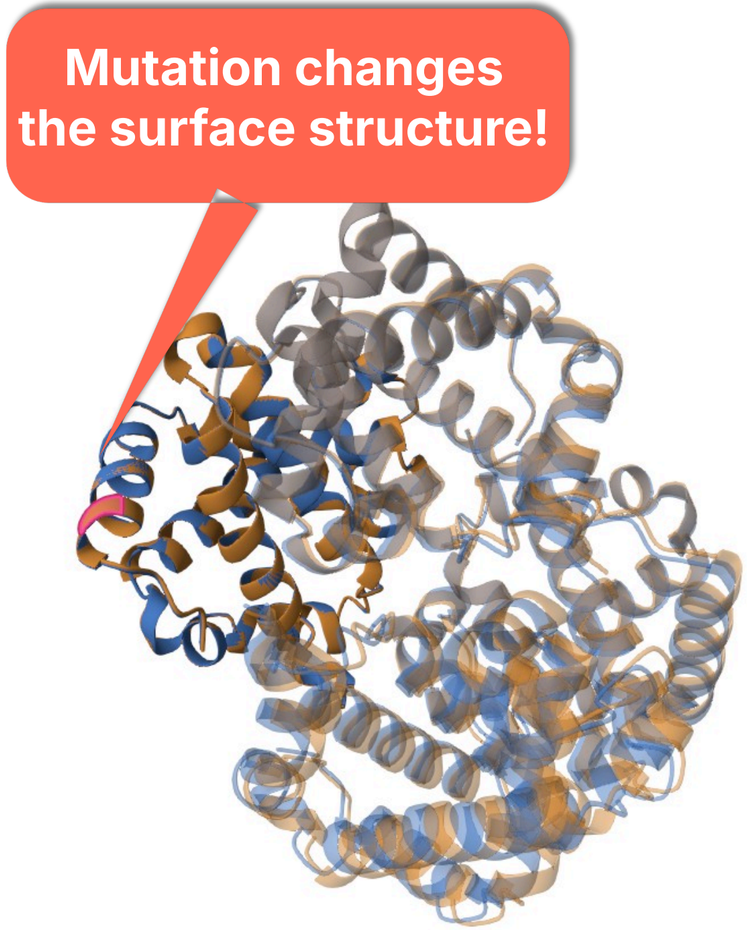
Zoom-In Feature

A feature that focuses on a specific amino acid in the amino acid chain of interest

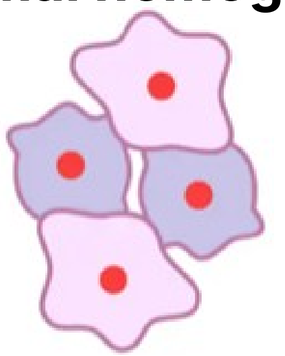


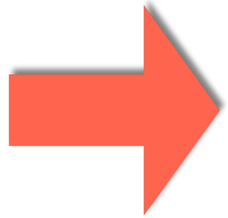
The Cause of the Disease

The surface structure, which is critical for protein interactions, changes, causing hemoglobin to clump together when it normally would not



Normal hemoglobin





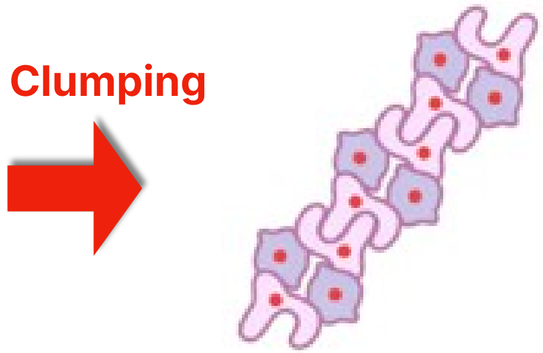
Normal red cell



Sickle cell Hemoglobin



Clumping



Sickle cell



Exercise Summary

Exercise summary

Using AlphaFold, we compared the structures of a normal protein and a mutated protein and visualized the cause of a disease

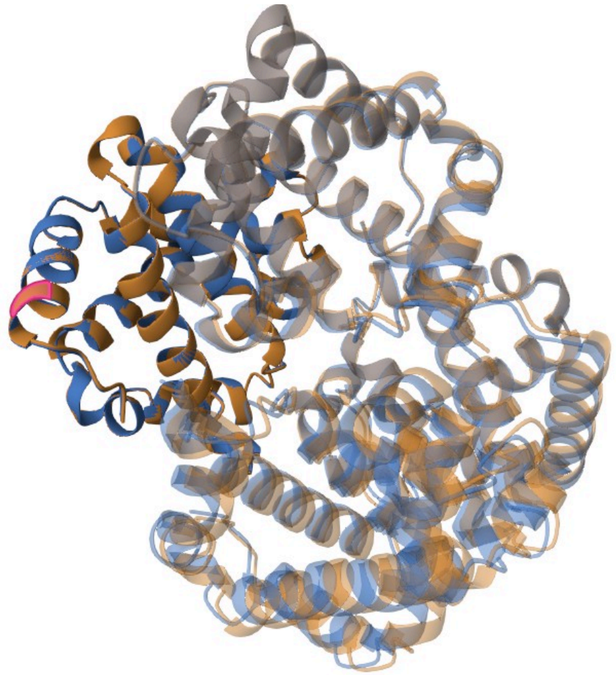
Normal sequence

MVHLTP**E**EKSAVTAL…
7

Disease-causing sequence

MVHLTP**V**EKSAVTAL…
7

AlphaFold



Contents

1 The Central Dogma (Flow of Genetic Information)

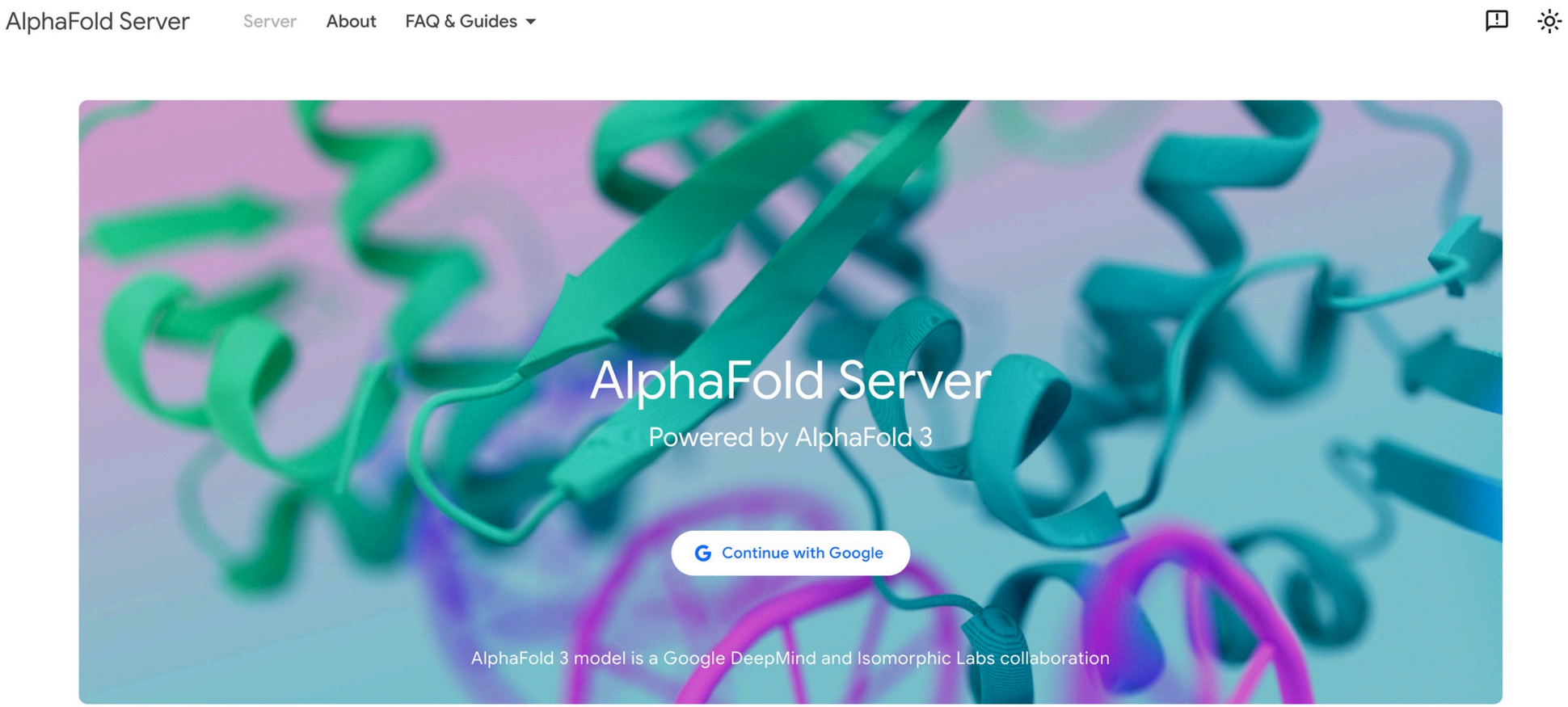
2 Reproducing the Central Dogma in Python

3 Why Protein Structure Matters, and AlphaFold

4 Applications in Medicine and Pharmacology

Alphafold 3

The latest model, AlphaFold 3, predicts not only 3D structures but also binding between proteins and molecules with high accuracy



Alphafold 3

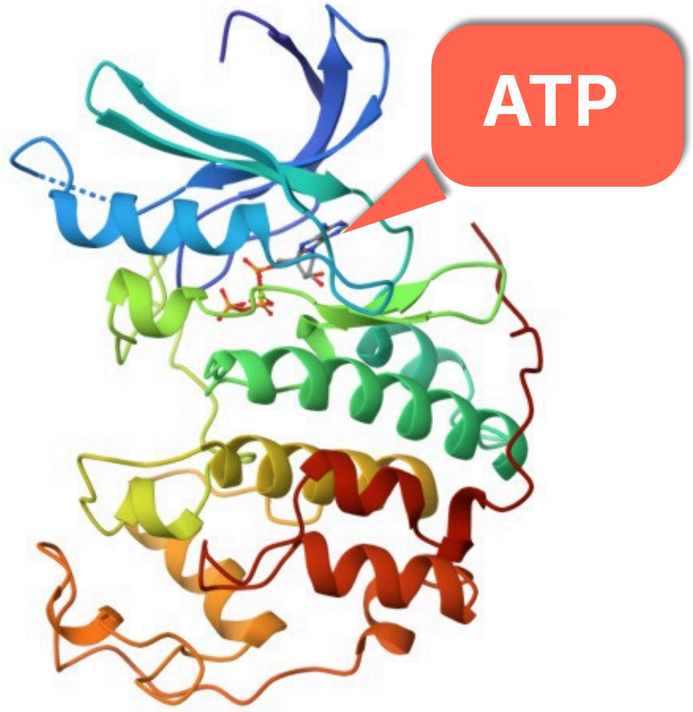
The latest model, AlphaFold 3, predicts not only 3D structures but also binding between proteins and molecules with high accuracy



CDK2

+ ATP

Predict



How is it actually used in medicine and pharmacology?

Impact on Medicine & Pharmacology

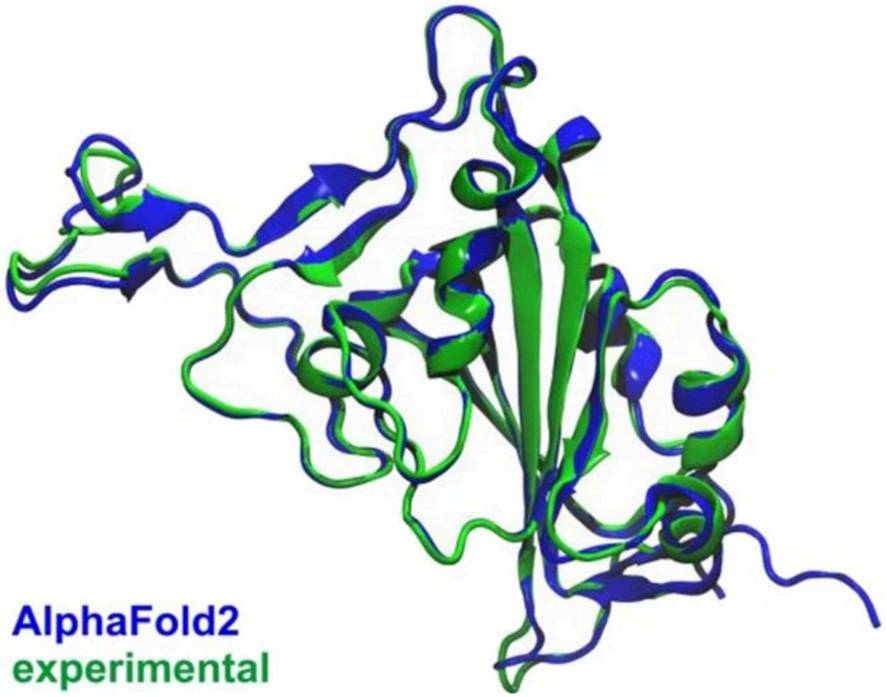
AlphaFold is helping medicine and pharmacology in the following ways

1. Uncovering disease mechanisms
2. Speeding up drug development
3. Personalized medicine

Impact on Medicine & Pharmacology

1. Uncovering disease mechanisms

Proteins that cause diseases left unexplained because of high cost and other reasons can now be predicted with AlphaFold, making the mechanisms understandable



Coronavirus protein

Coronavirus protein

+

Human protein

Using AlphaFold

AlphaFold

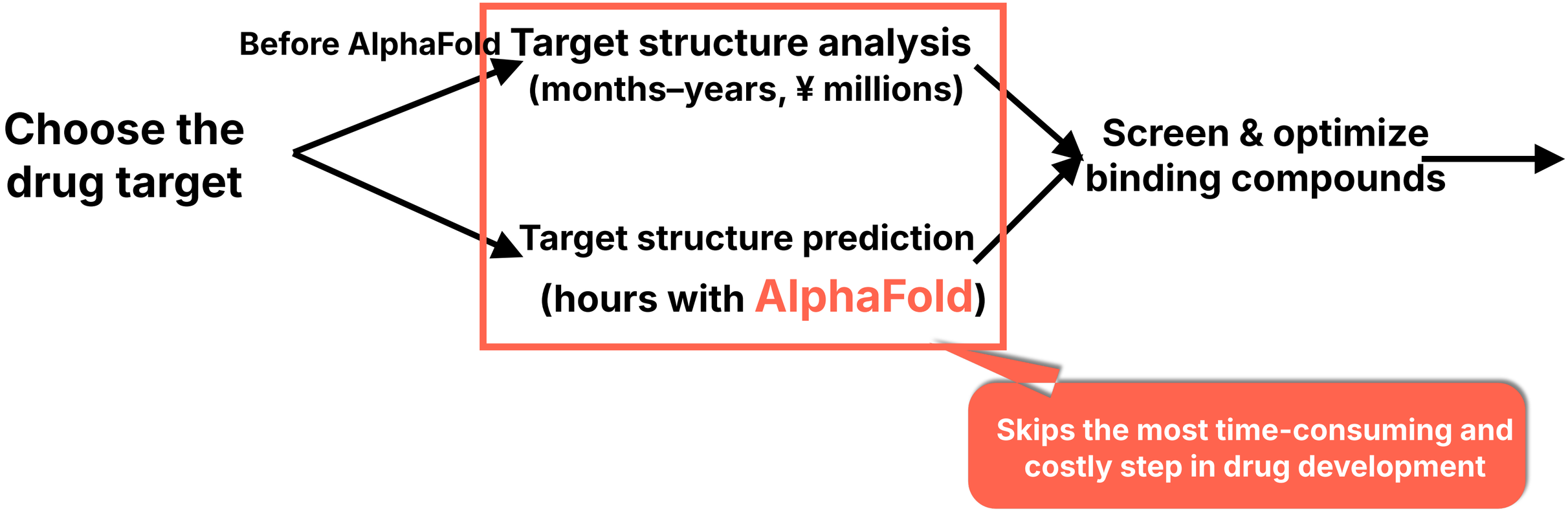
How do they interact to cause infection?

Helps uncover the mechanism

Impact on Medicine & Pharmacology

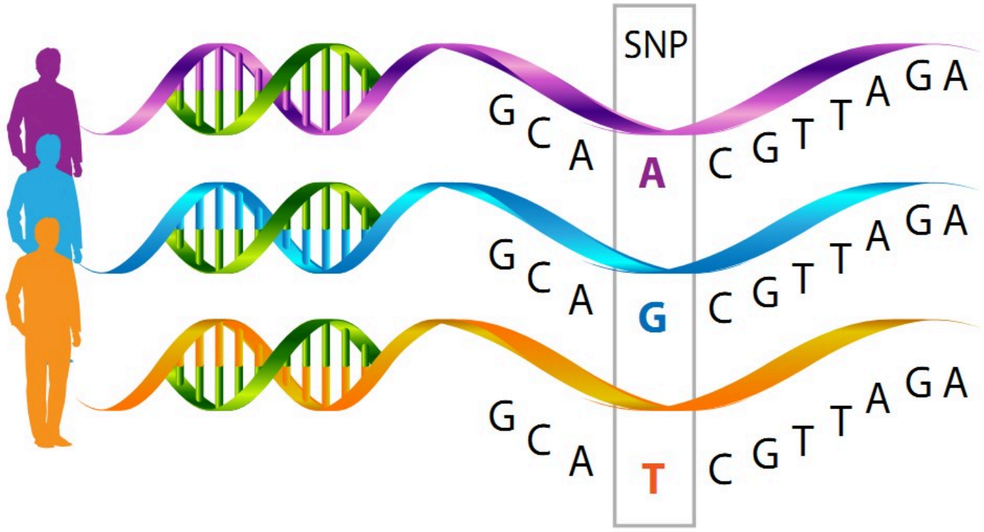
2. Faster Drug Development

New drugs that bind to disease-related proteins and suppress symptoms can be developed at dramatically lower cost (cheaper and faster)



3. Personalized Medicine

In the future, it may become a core technology of personalized medicine, which uses genome data to propose treatments tailored to individual differences



DNA sequences differ by about 0.1% between individuals

Using AlphaFold

AlphaFold

We can get data on the proteins working in each person and provide treatment tailored to it

Course Summary

- The essence of the central dogma, the mechanism by which DNA shapes life, is simple enough to be expressed as a program

- Knowing the 3D structure of proteins is important in biology, medicine, pharmacology, and other fields, and AlphaFold contributes greatly to these fields